

Motionator

User Guide

Animate SVG without code — 4K video, Lottie and animated SVG

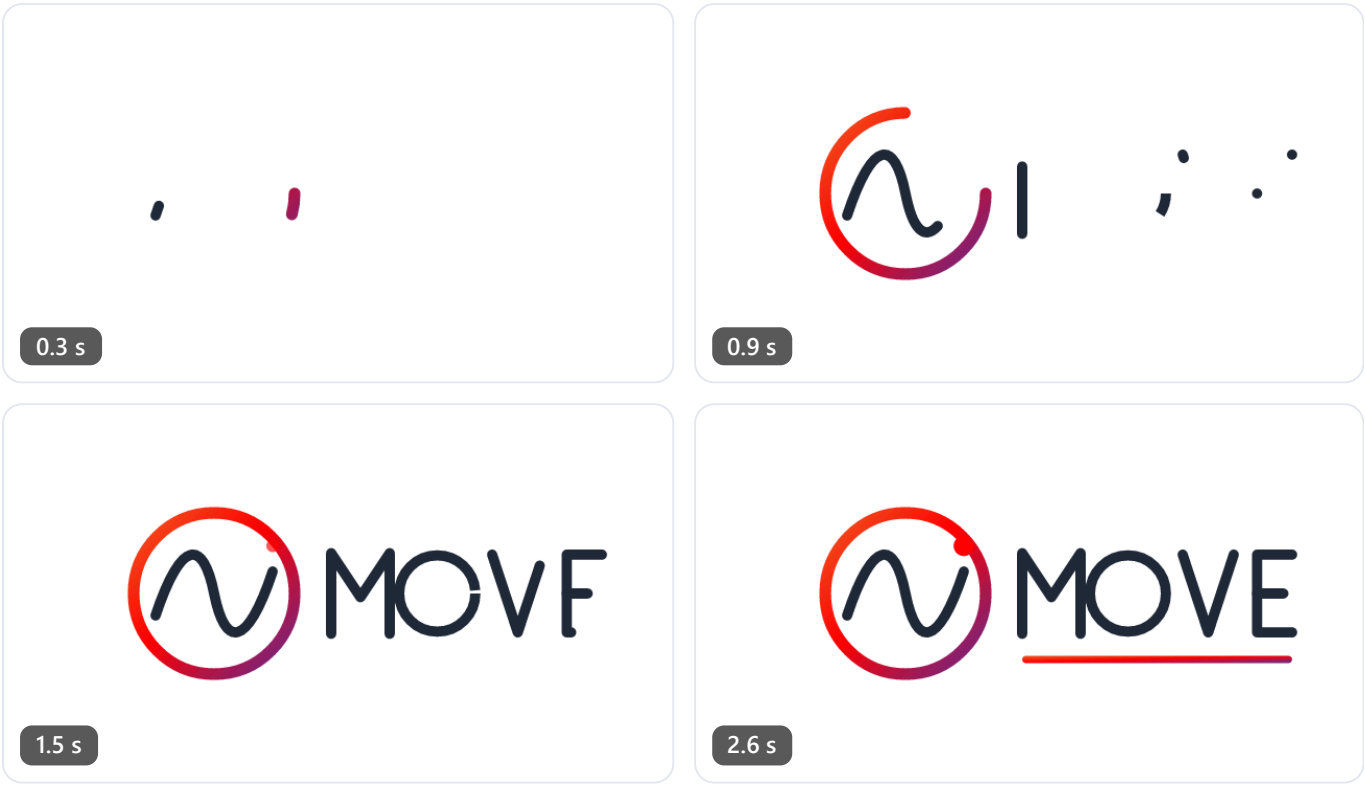
Version 0.1.0 · Online version: s4rt4.github.io/motionator-docs

Contents

1. Installation
2. Quick start: your first animation
3. Preparing SVG files
4. The workspace at a glance
5. Workspaces and docking panels
6. Layers panel
7. Composition panel
8. Properties panel
9. Presets
10. Keyframes and easing
11. Graph Editor
12. Motion paths
13. Trim Paths: drawing lines on
14. Effects and what each export supports
15. Saving your own presets
16. Audio
17. Video export: Render Queue
18. Lottie export
19. Animated SVG export
20. Projects, auto-save and recovery
21. Tutorial: logo reveal with sound
22. Tutorial: seamless 4K stock background
23. Tutorial: animated icon for a website
24. Keyboard shortcuts
25. Troubleshooting and FAQ
26. About and licences

Welcome

This guide explains every part of Motionator, from the first import to a finished 4K clip, a Lottie file or an animated SVG for your website. Each chapter has real screenshots, and the tutorials use sample files you can download and follow step by step.



A logo drawn on with Trim Paths, staggered letters and a wipe (sample: logo.motionator).

What you can make

Stock footage

Seamless 4K backgrounds in H.264 or ProRes, checked for loops and for the usual stock-site rules.

Logo animation

Draw-on strokes, pops, wipes and colour flashes, with a soundtrack.

Web animation

Lottie JSON for lottie-web and apps, or one self-contained animated SVG.

Your own presets

Turn any keyframed layer into a preset and reuse it in every project.

How Motionator works

You prepare artwork in Illustrator, Inkscape, Figma or any vector editor and save it as SVG. Every group or path with a name (an `id` in the SVG) becomes a **layer** you can animate. You then animate layers in two ways, which can be combined:

- **Presets**: ready-made motions (Fade In, Bounce, Orbit, Draw On…) applied with a double-click. Ideal for fast, consistent results.
- **Keyframes**: set values at points in time, with easing, a graph editor and curved motion paths, like in After Effects.

The same renderer draws the preview and the exported video, so what you see is what you get. Lottie and animated SVG exports are checked against this renderer and each export shows a compatibility report before saving.

01 Installation

Install once, then open SVG files or projects straight from Explorer.

Requirements

ITEM	REQUIREMENT
System	Windows 10 or 11, 64-bit
Memory	8 GB RAM; 16 GB recommended for 4K export with motion blur
Display	1600×900 or larger
WebView2	Included in Windows 11; the installer adds it on Windows 10 if missing
FFmpeg	Bundled with the installer (used for video and audio export)

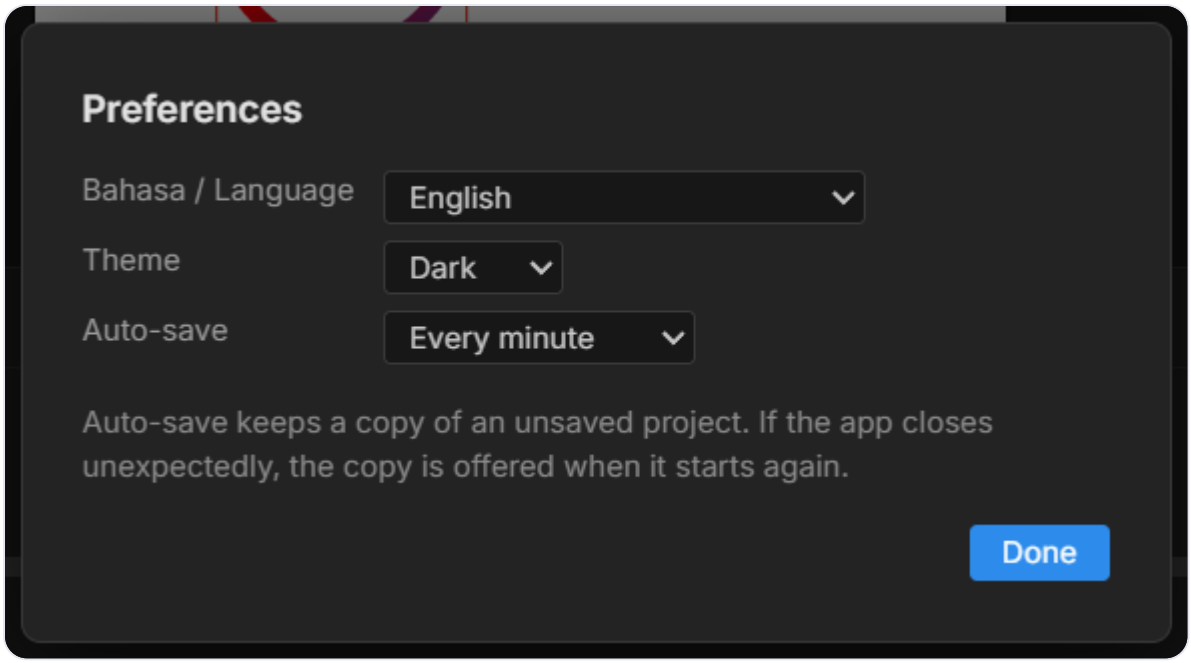
Installing

1. Run `Motionator_0.1.0_x64-setup.exe`. Choose **English** or **Bahasa Indonesia** for the installer.
2. Follow the pages and click **Install**. Motionator installs for your user account only, so no administrator rights are needed.
3. Optionally create a desktop shortcut, then start Motionator from the last page or the Start menu.

The installer registers `.motionator` project files: double-clicking one in Explorer opens it in Motionator. You can also right-click an `.svg` file and choose **Open with ▸ Motionator** to import it straight away.

First launch and language

Motionator follows the Windows display language: Indonesian on an Indonesian Windows, English otherwise. Change it any time in **Edit ▸ Preferences** (*Bahasa / Language*); the whole interface switches at once, including preset descriptions and export reports.



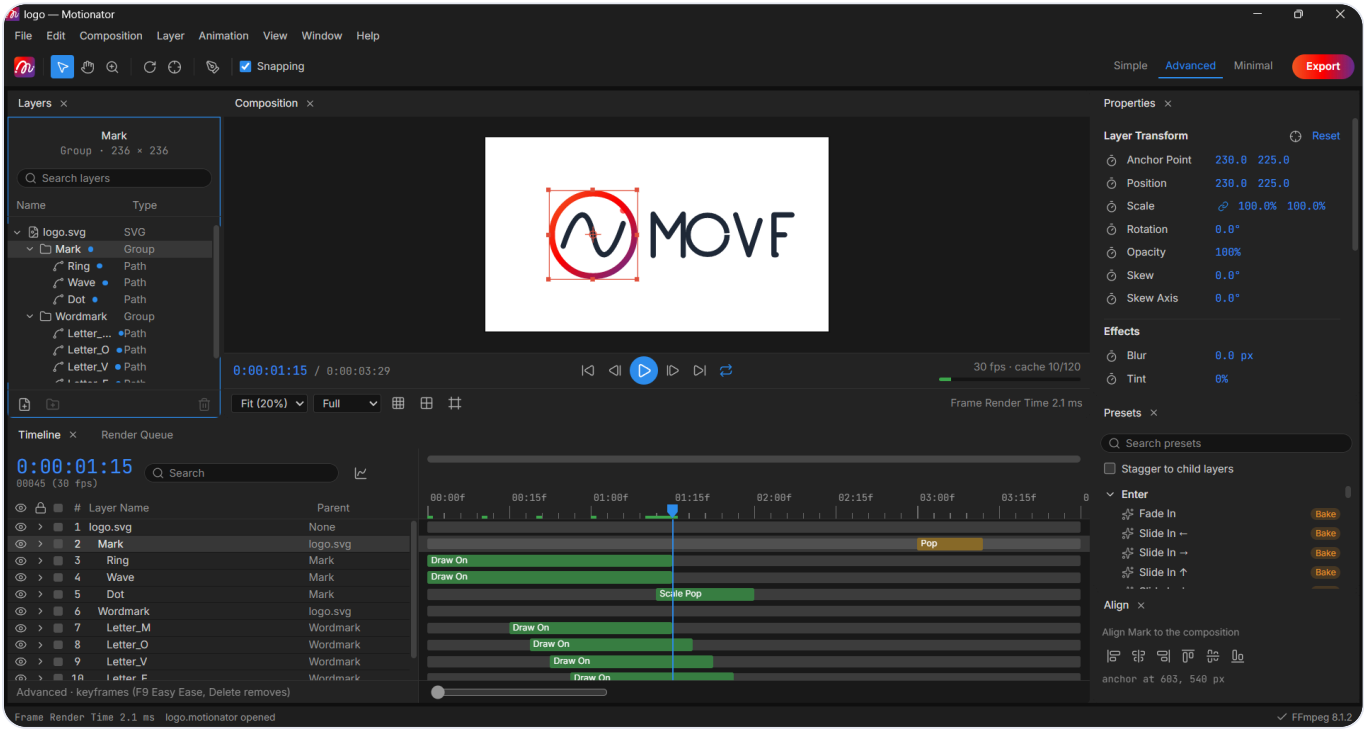
Preferences: language, theme (Dark or Darker) and auto-save interval.

> **NOTE**
> The status bar shows **FFmpeg** with a check mark when video export is ready. If it shows a warning, reinstall Motionator; video export needs FFmpeg, everything else works without it.

02 Quick start: your first animation

Five minutes from an SVG to a finished MP4.

1. **Import an SVG.** Press `Ctrl` + `I`, choose **File ▸ Import SVG...**, or simply drag an `.svg` file onto the window. Try `samples/logo.svg`.
2. **Select a layer.** Click it in the **Layers** panel or directly on the canvas with the Selection tool `V`.
3. **Apply a preset.** In the **Presets** panel, hover a preset to preview it, then **double-click** it. Enter presets start at the playhead (the blue line in the timeline).
4. **Play it.** Press `Space` or the round play button under the canvas. The green bar on the ruler shows frames already cached for smooth playback.
5. **Adjust timing.** In the **Timeline**, drag a preset bar to move it and drag its right edge to change its duration.
6. **Export.** Click the red **Export** button (or `Ctrl` + `M`), choose a format and size in the Render Queue, pick the output file, then press **Render**.
7. **Save the project** with `Ctrl` + `S`. The `.motionator` file contains your SVG, so it can be moved and shared as one file.



The logo sample at 1.5 s: layers on the left, the canvas in the middle, presets as bars in the timeline.

TIP

Everything you do can be undone with `Ctrl` + `Z` and redone with `Ctrl` + `Shift` + `Z`. A whole drag is one undo step.

03 Preparing SVG files

Good layer names in your design tool make animation easy.

Named elements become layers

Motionator makes a layer for every group and path that has an `id`. Elements without an id stay part of their parent layer, so they move together. Nesting is kept: a named group inside another named group becomes a child layer, and moving the parent moves the child.

TOOL	HOW TO GIVE ELEMENTS IDS
Adobe Illustrator	Name layers and sub-layers in the Layers panel. Export with **File** ▸ **Export** ▸ **Export As...** ▸ **SVG** and *Object IDs: Layer Names*. Illustrator writes spaces as `_x20_`; Motionator shows them as spaces again.
Inkscape	Set the **ID** in Object Properties (`Ctrl` + `Shift` + `O`), or rename layers. Save as *Plain SVG* or *Inkscape SVG*.
Figma	Layer names become ids when you export a frame as SVG with *Include "id" attribute* turned on.
Any editor	Add `id="Name"` to the `<g>` or `<path>` elements in a text editor.

Checklist before importing

- **Outline text** (Create Outlines / Object to Path). Live text is not drawn.
- **Use strokes for line art** you want to draw on with Trim Paths. Filled shapes cannot be drawn on.
- **Group what moves together** and name the group, e.g. a wheel with its spokes.
- **Avoid filters and raster images** if you also need Lottie: they render in video but cannot be exported to Lottie.
- **Keep ids unique.** Duplicate ids are reported; only the first becomes a layer.

Import warnings

After an import the status bar shows how many layers were found. If something needs attention, a warning appears there; click it to see the list. Warnings never stop the import.

WARNING	WHAT TO DO
Text not outlined	Convert text to outlines and import again.
Filters (blur, shadow...)	Fine for video; slows 4K rendering and is left out of Lottie.
Patterns / blend modes	Shown in video; Lottie support is limited.
Raster images	Blurry when scaled up; makes Lottie files larger.
Duplicate ids	Rename them in your editor so each part can be animated.
No ids	The whole SVG becomes one layer; name your layers.

A newly imported SVG is centred in the composition at its own size, scaled down only if it is larger than the composition. Each layer's anchor point starts at the centre of its box.

04 The workspace at a glance

An After Effects–style layout: panels around a central canvas.

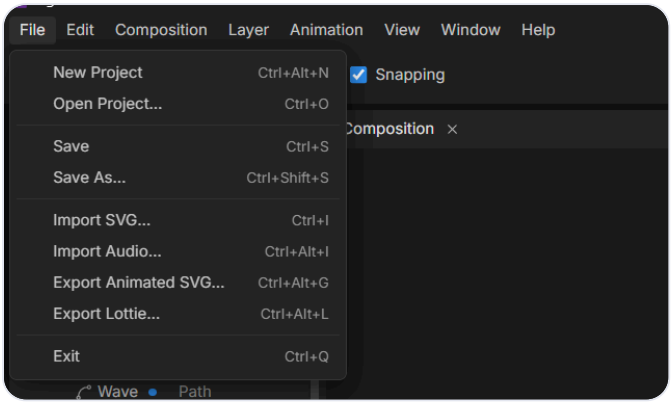


The main window in the Advanced workspace.

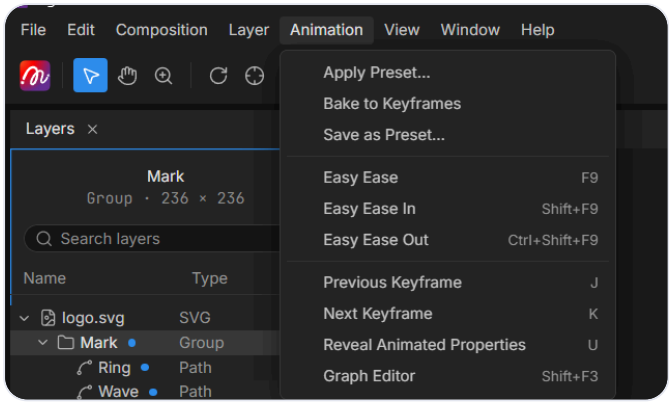
1. **Menu bar**: every command, with its shortcut.
2. **Toolbar**: Selection, Hand, Zoom, Rotation, Anchor Point and Pen tools, and Snapping.
3. **Workspaces**: Simple, Advanced and Minimal.
4. **Export**: adds the composition to the Render Queue.
5. **Layers**: the layer tree of your SVGs.
6. **Composition**: the canvas, where you select and move layers.
7. **Transport**: play, step, loop, current time and cache.
8. **Properties**: transform, effects, Trim Paths and presets of the selected layer.
9. **Timeline**: preset bars, keyframes and the Graph Editor; the Render Queue is its neighbour tab.
10. **Presets**: the preset library with search and live previews.
11. **Align**: align the selected layer to the composition.
12. **Status bar**: messages, import warnings and the FFmpeg check.

Menus

The menus list every command with its keyboard shortcut. The File menu holds projects, imports and the web exports; Animation holds preset and keyframe commands; Window shows and arranges panels.



File menu



Animation menu

Toolbar and tools



The toolbar: logo, tools, Snapping, workspaces and Export.

TOOL	KEY	USE
Selection	`V`	Click a layer on the canvas to select it; drag to move it. Hold `Shift` to move only horizontally or vertically.
Hand	`H`	Drag to pan the canvas. Holding `Space` gives the Hand tool temporarily.
Zoom	`Z`	Click to zoom in at that point; `Alt`-click zooms out.
Rotation	`W`	Drag around the selected layer to rotate it about its anchor point; `Shift` snaps to 45°.
Anchor Point	`Y`	Drag to move the anchor point (the pivot for rotation and scale) without moving the layer.

TOOL	KEY	USE
Pen	`G`	Shows the motion path; click a position keyframe to switch it between a smooth curve and a sharp corner.

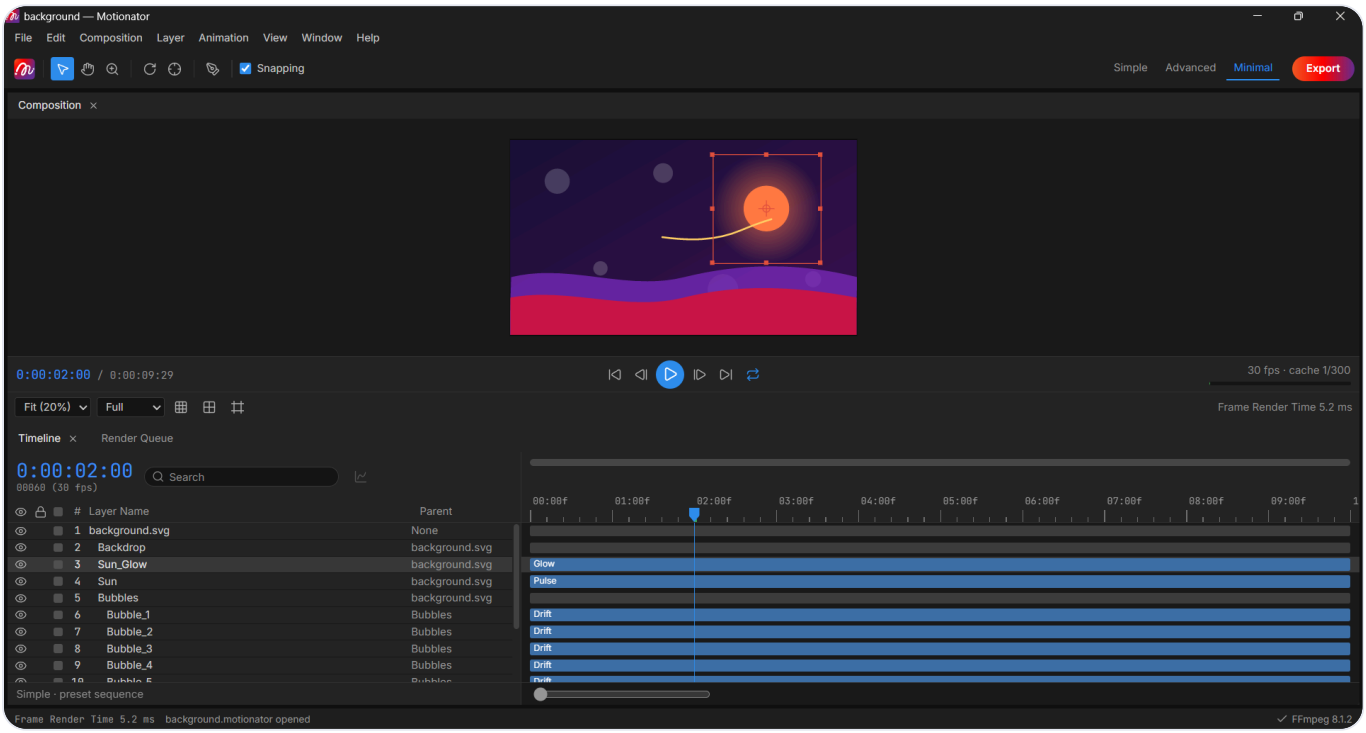
With the mouse wheel over the canvas: scroll pans, `Shift` +scroll pans sideways, and `Ctrl` +scroll or `Alt` +scroll zooms around the pointer.

05 Workspaces and docking panels

Arrange panels the way you work, and keep several layouts.

Simple, Advanced and Minimal

WORKSPACE	KEY	FOR
Simple	`Shift` + `F10`	Working with presets: the timeline shows one row per layer with preset bars.
Advanced	`Shift` + `F11`	Keyframes: layers twirl open to show every property, keyframes and the Graph Editor.
Minimal	`Shift` + `F12`	Reviewing: just the canvas and the timeline.

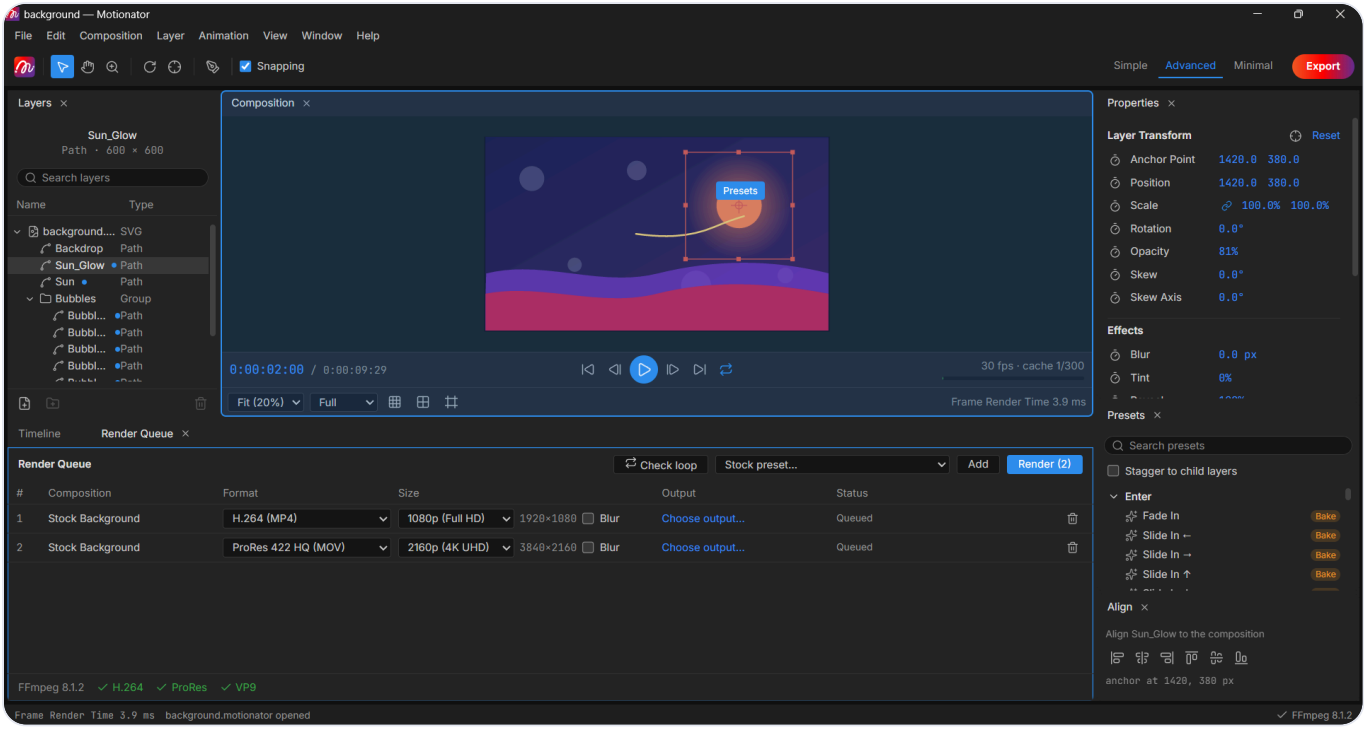


The Minimal workspace: more room for the canvas.

Moving panels

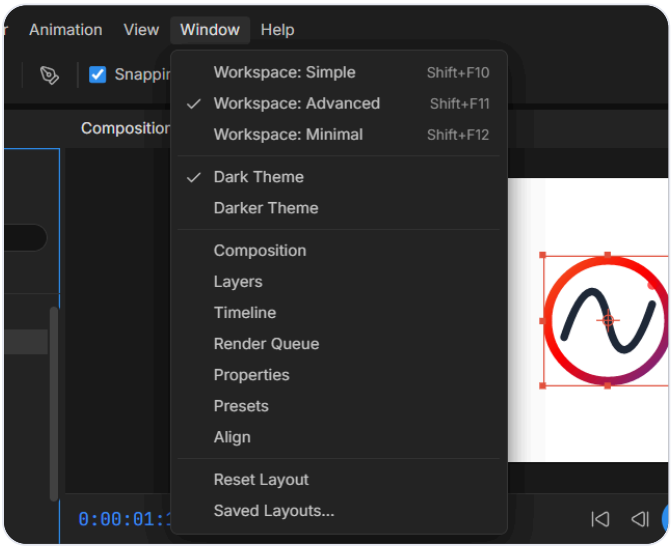
Drag a panel by its tab. A blue overlay shows where it will land: the outer quarter of a panel group docks it on that side, the middle adds it as a tab. Release to drop it there, or press `Esc` to cancel.

Drag the gaps between panels to resize them.

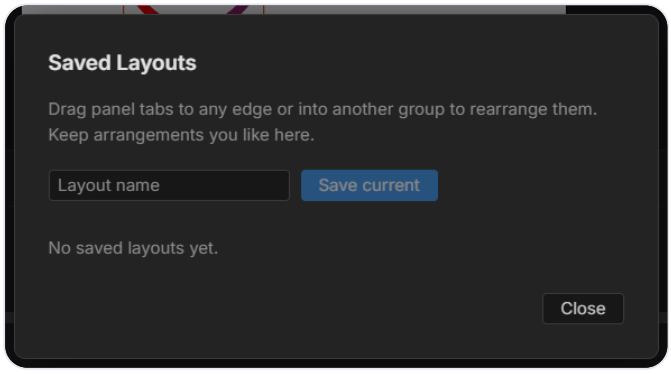


Dragging the Presets tab: the blue overlay marks the drop zone (here the middle of the Composition group, so it would become a tab).

Close a panel with the × on its tab and bring it back from the **Window** menu. Each workspace remembers its own arrangement. **Window ▸ Reset Layout** restores the default, and **Window ▸ Saved Layouts...** keeps arrangements under a name.



The Window menu: workspaces, themes, panels and layouts.

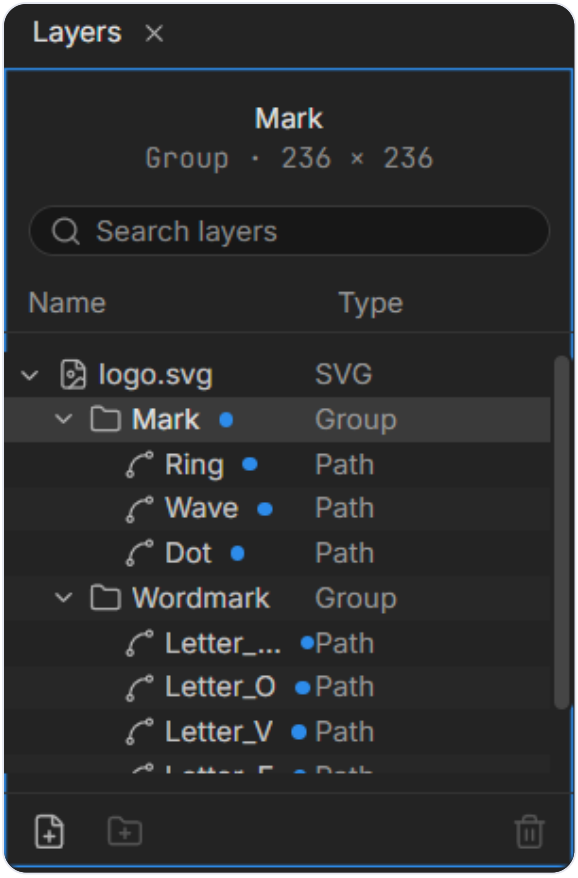


Saved Layouts: save the current arrangement, apply or delete saved ones.

TIP

Themes: **Window** ▸ **Dark Theme** or **Darker Theme**. The choice is remembered.

06 Layers panel

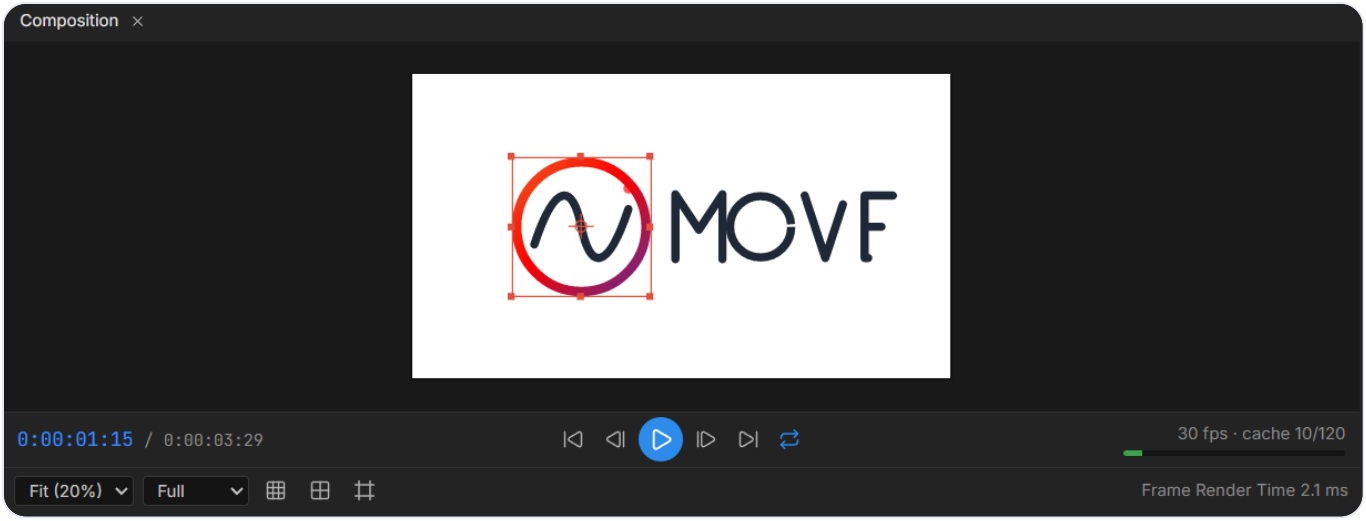


Layers panel with the logo sample: SVG root, groups and paths. A dot marks animated layers.

- **Header**: the selected layer's name, kind, size and SVG id.
- **Search**: type to filter layers by name.
- **Kinds**: *SVG* is the imported file (the root layer), *Group* and *Path* are named elements inside it. Arrows fold groups.
- **Blue dot**: the layer has presets or keyframes.
- **Buttons**: import another SVG, and delete the selected SVG (only whole SVG files can be deleted; their parts belong to the file).

A composition can hold several SVG files; each becomes its own root layer. Later imports appear on top.

07 Composition panel



The canvas with the selection box, anchor point (crosshair) and the bar of controls under it.

Canvas

The composition is the white (or chequered, when transparent) rectangle; the dark area around it is the pasteboard. The selected layer shows a red box that follows its rotation, eight handles and its anchor point. Hovering a layer outlines it in blue.

Playback controls

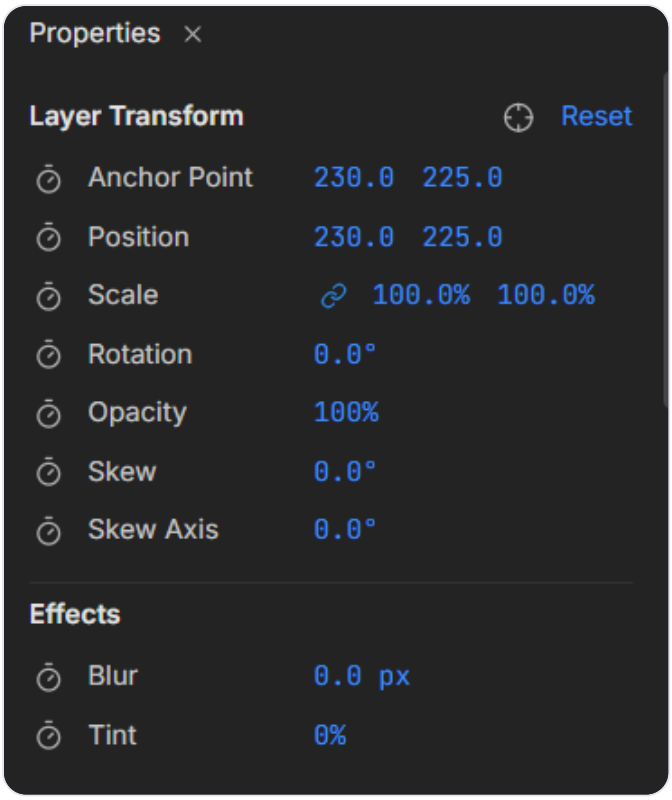
CONTROL	KEY	ACTION
Current time / duration		Timecode in hours:minutes:seconds:frames.
Go to start / end	`Home` `End`	
Previous / next frame	`Page Up` `Page Down`	Also the arrow keys; with `Shift` they move 10 frames.
Play / Pause	`Space`	Plays in real time with sound (if the project has audio).
Loop		Repeat playback from the start.
Cache		How many frames are ready; cached frames play smoothly.

Viewer options

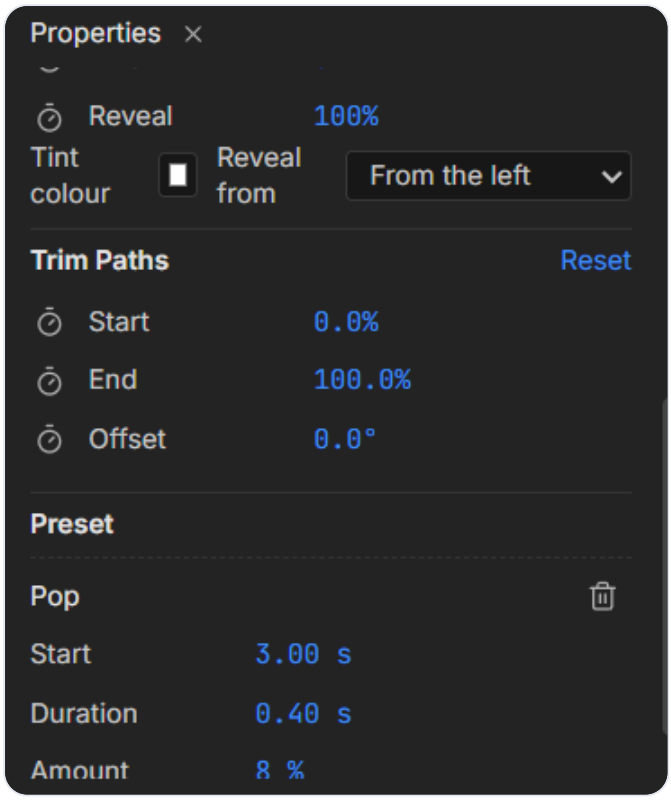
- **Zoom**: Fit, or a fixed percentage. Keys: `.` zoom in, `,` zoom out, `/` 100 %, `Shift`+`/` fit.
- **Resolution**: Full, Half, Third, Quarter. Lower resolutions preview faster on large compositions.
- **Grid** `Ctrl` + `'`, **transparency grid** and **safe areas** (action and title safe).
- **Frame Render Time**: how long the last frame took to draw.

08 Properties panel

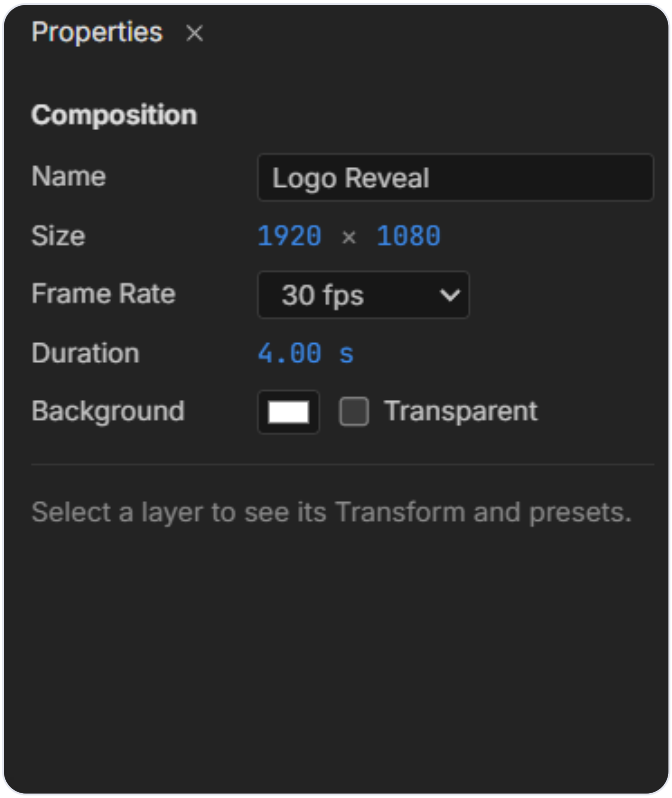
Exact values for the selected layer, or the composition settings when nothing is selected.



Layer Transform and Effects.



Further down: Trim Paths and the presets on this layer.



With nothing selected: the composition settings.

Editing values

Blue numbers can be **dragged** left and right to scrub the value, or **clicked** to type a number (press `Enter` to confirm, `Esc` to cancel). The clock icon is the **stopwatch**: turn it on to animate the property with keyframes (see Keyframes).

Layer Transform

PROPERTY	MEANING
Anchor Point	The pivot for rotation, scale and skew, in the parent's coordinates. The crosshair button centres it on the layer.
Position	Where the anchor point sits (x, y).
Scale	Width and height in percent. The chain button keeps the proportions.
Rotation	Degrees, clockwise.
Opacity	0–100 %.

PROPERTY	MEANING
Skew / Skew Axis	Slants the layer along an axis (degrees), like in After Effects.

Reset returns the transform to the rest pose of the SVG (effects are kept). **Layer ▸ Center Anchor Point** `Ctrl` + `Alt` + `Home` and **Layer ▸ Reset Transform** do the same from the menu.

Effects

EFFECT	MEANING
Blur	Gaussian blur radius in the layer's own pixels (it grows with the layer's scale).
Tint	How much of the tint colour replaces the layer's colours (0–100 %). Pick the colour with **Tint colour**.
Reveal	The visible part of the layer's box (100 % = all), wiping in from the side chosen in **Reveal from**: left, right, top, bottom, iris (circle) or centre.

Composition settings

Click an empty part of the canvas (or press `Ctrl` + `K`) to edit the composition: **name**, **size** in pixels, **frame rate** (23.976 to 60 fps), **duration** in seconds and the **background** colour. Tick **Transparent** for a transparent background, needed for ProRes 4444 or VP9 with alpha.

Align panel

The six buttons align the selected layer's box to the left, centre or right and to the top, middle or bottom of the composition. If Position is keyframed, the change is keyed at the playhead.

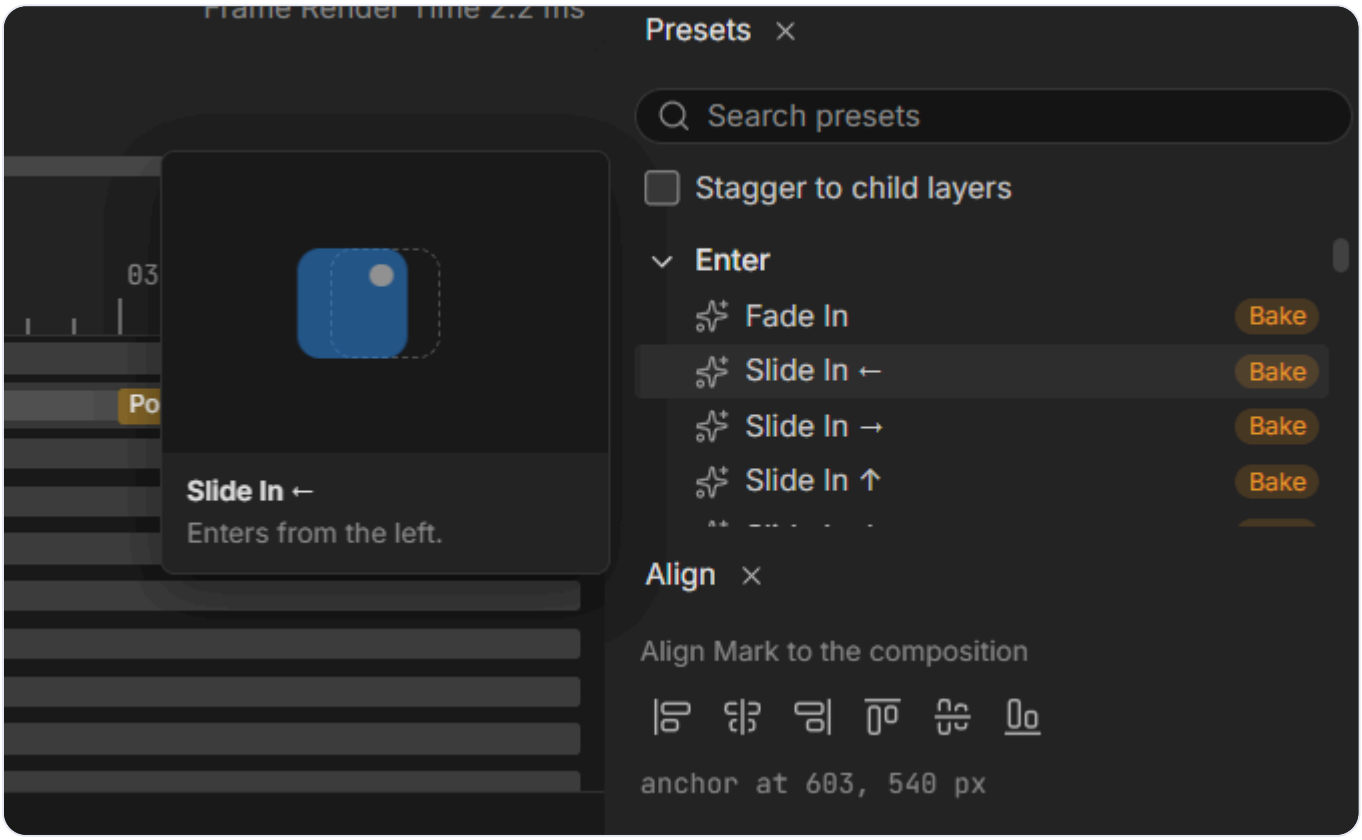
09 Presets

82 ready-made motions in four categories, applied with a double-click.

The four categories

CATEGORY	WHAT IT DOES	WHERE IT IS PLACED
Enter	Brings the layer in: it ends in the layer's rest pose.	Starts at the playhead.
Exit	Takes the layer out: it starts from the rest pose.	Ends at the end of the composition.
Loop	Repeats for as long as the bar lasts and loops seamlessly.	Covers the whole composition.
Emphasis	A short accent that returns to where it started.	Starts at the playhead.

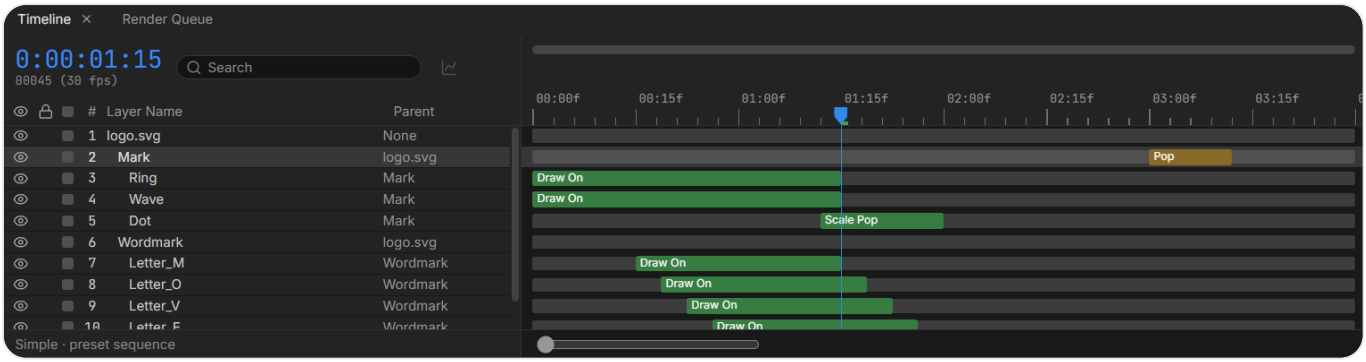
Applying a preset



Hovering a preset plays a live preview of it on a sample shape.

1. Select a layer (in the Layers panel, the timeline or on the canvas).
2. For Enter and Emphasis presets, move the playhead to where the motion should start.
3. Find the preset: type in **Search presets** or open a category. Hover to preview.
4. Double-click the preset. It appears as a bar in the timeline and in the **Preset** section of the Properties panel.

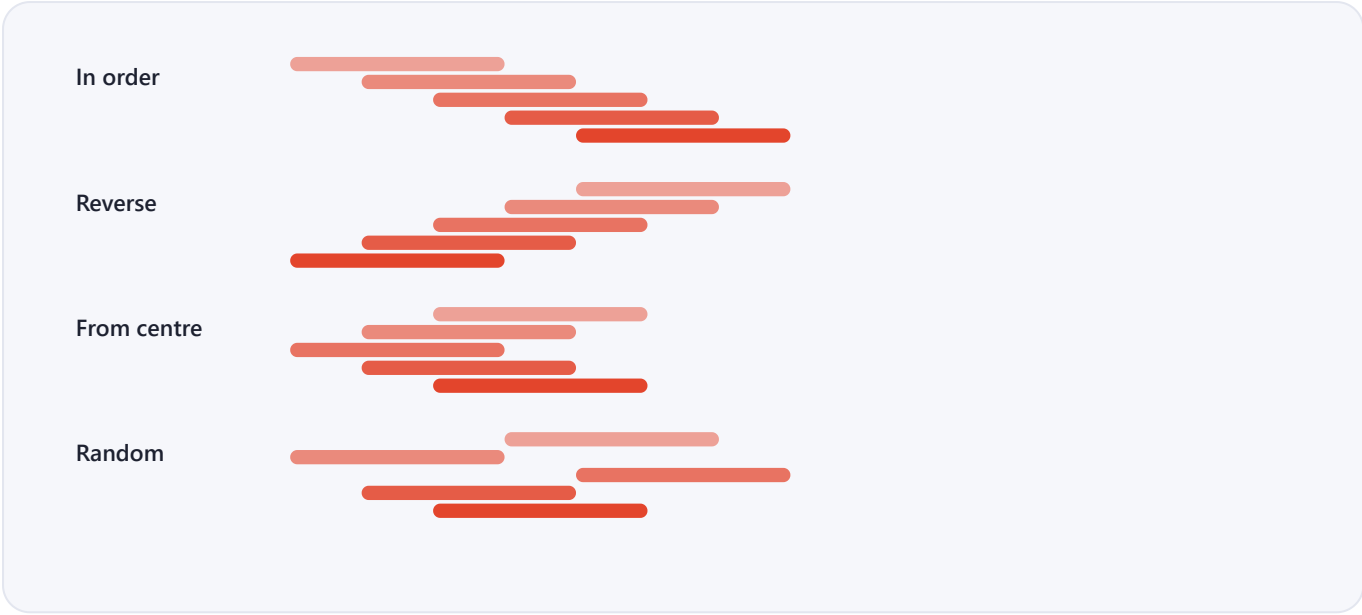
A layer can have any number of presets. Select a bar in the timeline to edit its **Start**, **Duration** and parameters (distance, amount, period, direction…) in the Properties panel, or delete it with the trash icon or `Delete`.



The Simple timeline: one row per layer, presets as coloured bars. Drag a bar to move it, drag its edges to change its length.

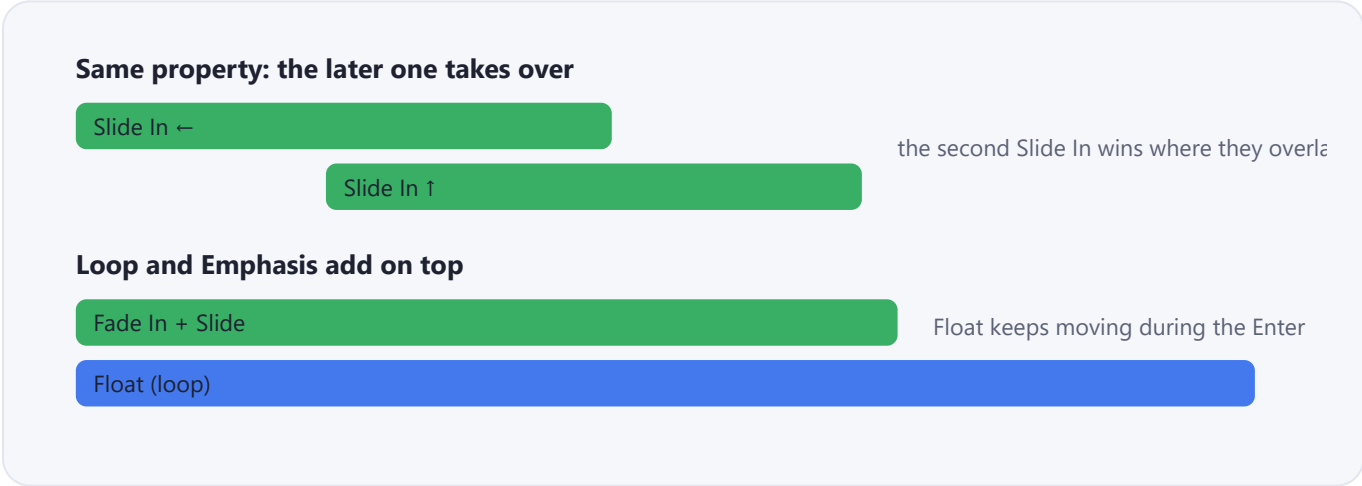
Stagger: animate children one after another

Tick **Stagger to child layers** at the top of the Presets panel, select a group and double-click a preset. Instead of the group, each child layer gets the preset, each starting a little later. Set the delay between them (seconds) and the order:



The letters of the logo sample use stagger: *Pop Up* on the group `Wordmark` with 0.08 s between letters. Random order is seeded, so it is the same on every render.

Combining presets



- **Enter and Exit** presets *override*: where two of them change the same property at the same time, the one that starts later wins.
- **Loop and Emphasis** presets are *added* on top of everything else, so a layer can float while it slides in.
- **Keyframes** are the base: presets work relative to the keyframed value.
- Presets that rotate or scale (Swing, Spin, Pop…) turn around the **anchor point**; move it first if the pivot matters.

Bake to Keyframes

Animation ▸ Bake to Keyframes turns the presets of the selected layer into ordinary keyframes, so you can fine-tune them in the Graph Editor. The badge next to each preset (*Bake*) means it is exported to Lottie in this baked form.

All presets

CATEGORY	PRESET	DESCRIPTION
Enter	**Fade In**	Fades in from transparent.
Enter	**Slide In ←**	Enters from the left.
Enter	**Slide In →**	Enters from the right.
Enter	**Slide In ↑**	Rises from below.
Enter	**Slide In ↓**	Comes down from above.
Enter	**Scale Pop**	Grows from zero with a slight overshoot.
Enter	**Rotate In**	Rotates in while fading in.
Enter	**Bounce In**	Drops from above and bounces.
Enter	**Elastic Pop**	Grows with an elastic wobble.
Enter	**Drop**	Falls from above, accelerating.
Enter	**Zoom In**	Comes in large and shrinks to size.
Enter	**Fly In**	Flies in diagonally.
Enter	**Spiral In**	Spins while growing from zero.
Enter	**Roll In**	Rolls in from the left.
Enter	**Flip In X**	Flips in around the vertical axis (3D look).
Enter	**Flip In Y**	Flips in around the horizontal axis (3D look).
Enter	**Pop Up**	Pops up with a slight overshoot.
Enter	**Swing In**	Swings in and settles (put the anchor on the hinge).

CATEGORY	PRESET	DESCRIPTION
Enter	**Stretch In**	Comes in stretched and springs into shape.
Enter	**Draw On**	Strokes draw on from the start of each path to its end (needs a stroke).
Enter	**Draw From Centre**	Strokes draw out from the middle of each path to both ends (needs a stroke).
Enter	**Blur In**	Comes into focus from a blur.
Enter	**Shear In**	Slides in slanted, then straightens.
Enter	**Wipe In**	Wipes on from one side.
Enter	**Iris In**	Opens from a small circle in the middle.
Enter	**Split Reveal**	Opens from the centre line out to both sides.
Exit	**Fade Out**	Fades out to transparent.
Exit	**Scale Out**	Shrinks until it disappears.
Exit	**Slide Out ←**	Exits to the left.
Exit	**Slide Out →**	Exits to the right.
Exit	**Slide Out ↑**	Exits upwards.
Exit	**Slide Out ↓**	Exits downwards.
Exit	**Rotate Out**	Rotates out while fading out.
Exit	**Bounce Out**	Hops up, then falls out.
Exit	**Zoom Out**	Grows while fading out.
Exit	**Fly Out**	Flies out diagonally.
Exit	**Spiral Out**	Spins while shrinking away.
Exit	**Roll Out**	Rolls out to the right.
Exit	**Flip Out X**	Flips out around the vertical axis.

CATEGORY	PRESET	DESCRIPTION
Exit	**Flip Out Y**	Flips out around the horizontal axis.
Exit	**Sink**	Sinks a little, shrinks and fades.
Exit	**Collapse**	Flattens to a line and disappears.
Exit	**Draw Off**	Strokes erase from the start of each path to its end (needs a stroke).
Exit	**Blur Out**	Blurs away while fading out.
Exit	**Wipe Out**	Wipes off towards one side.
Exit	**Iris Out**	Closes to a circle in the middle.
Loop	**Float**	Floats up and down continuously.
Loop	**Pulse**	Grows and shrinks rhythmically.
Loop	**Spin**	Spins continuously.
Loop	**Swing**	Swings like a pendulum (anchor = pivot).
Loop	**Breathe**	Slowly grows and dims, like breathing.
Loop	**Blink**	Blinks briefly at a regular interval.
Loop	**Orbit**	Moves in a circle, starting and ending at its own position.
Loop	**Figure 8**	Moves along a figure eight.
Loop	**Bounce**	Hops in place with squash and stretch.
Loop	**Wobble**	Wobbles and stretches gently, continuously.
Loop	**Glow**	Dims and brightens rhythmically.
Loop	**Drift**	Drifts slowly and irregularly, still seamless when looped.
Loop	**Hover**	Hovers up and down with a slight tilt.
Loop	**Tick**	Rotates in steps like a clock hand.

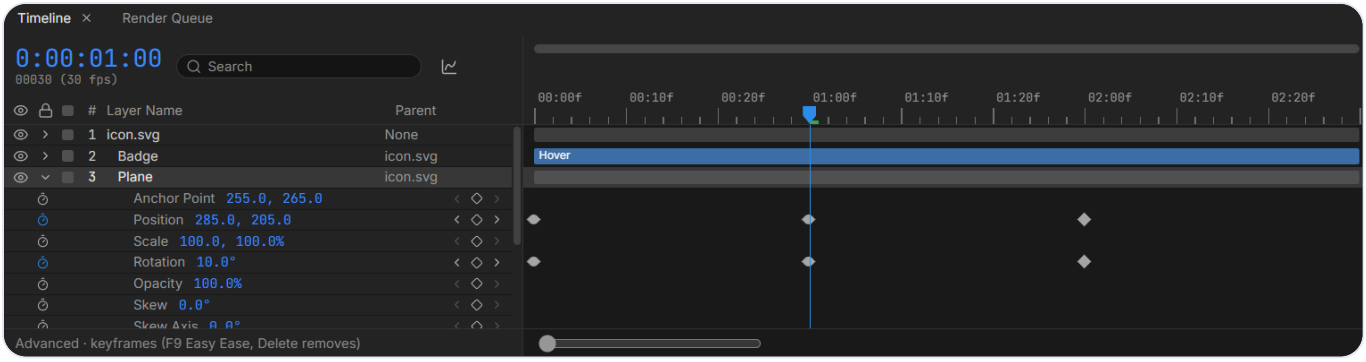
CATEGORY	PRESET	DESCRIPTION
Loop	**Trace**	A segment of the stroke travels along each path continuously (needs a stroke).
Loop	**Sway**	Sways from side to side like in the wind.
Loop	**Tint Pulse**	A colour pulses in and out continuously.
Emphasis	**Shake**	Shakes left and right, then settles.
Emphasis	**Heartbeat**	Two quick beats.
Emphasis	**Tada**	Shrinks, grows while wobbling, then settles.
Emphasis	**Jello**	Wobbles like jelly.
Emphasis	**Wiggle**	Smooth random motion (seeded, identical on every render).
Emphasis	**Flash**	Flashes a few times.
Emphasis	**Rubber Band**	Stretches and snaps back like rubber.
Emphasis	**Head Shake**	Shakes its head, like saying no.
Emphasis	**Nod**	Nods twice.
Emphasis	**Spin**	Spins full turns, then stops.
Emphasis	**Flip**	Flips over horizontally once.
Emphasis	**Pop**	Grows for a moment, then returns.
Emphasis	**Squash**	Squashes, then springs back.
Emphasis	**Nudge**	Nudges aside, then returns.
Emphasis	**Swing**	Swings a few times, then rests (anchor = pivot).
Emphasis	**Focus Pulse**	Blurs for a moment, then sharpens again.
Emphasis	**Lean**	Leans to one side, then springs back.
Emphasis	**Color Flash**	Flashes a colour for a moment, then returns.
Emphasis	**Jump**	Jumps twice, the second one lower.

TIP

Presets marked *needs a stroke* (Draw On, Draw From Centre, Draw Off, Trace) animate Trim Paths and only show on paths with a stroke.

10 Keyframes and easing

Full control: set values at moments in time and Motionator fills in the frames between.



The Advanced timeline: Plane twirled open, Position and Rotation have keyframes (diamonds).

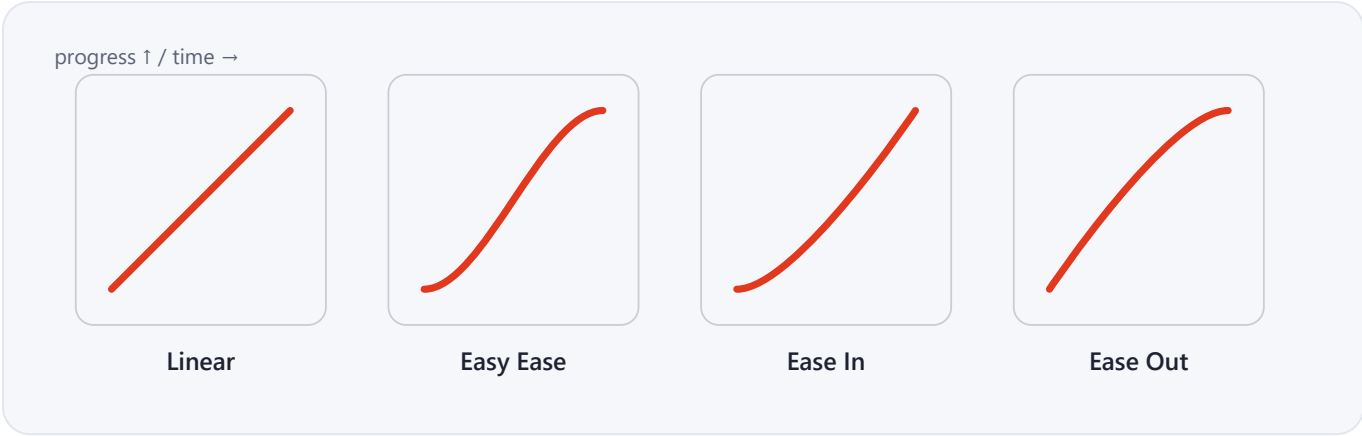
Making keyframes

1. Switch to the **Advanced** workspace `Shift` + `F11` and twirl a layer open with its arrow.
2. Click the **stopwatch** next to a property (here or in the Properties panel). A keyframe is made at the playhead with the current value.
3. Move the playhead and change the value, by dragging the number, typing, or moving the layer on the canvas. A new keyframe is added automatically.
4. Use the ◇ button to add or remove a keyframe at the playhead, and ‹ › to jump between keyframes (`J` / `K`).

- **Select** keyframes by clicking; `Shift`-click to select several.
- **Move** them by dragging; `Delete` removes them.
- Clicking the stopwatch again removes all keyframes of that property.
- `U` shows only the properties that are animated.

Easing

Easing controls how a value speeds up and slows down between keyframes. Select keyframes and press:



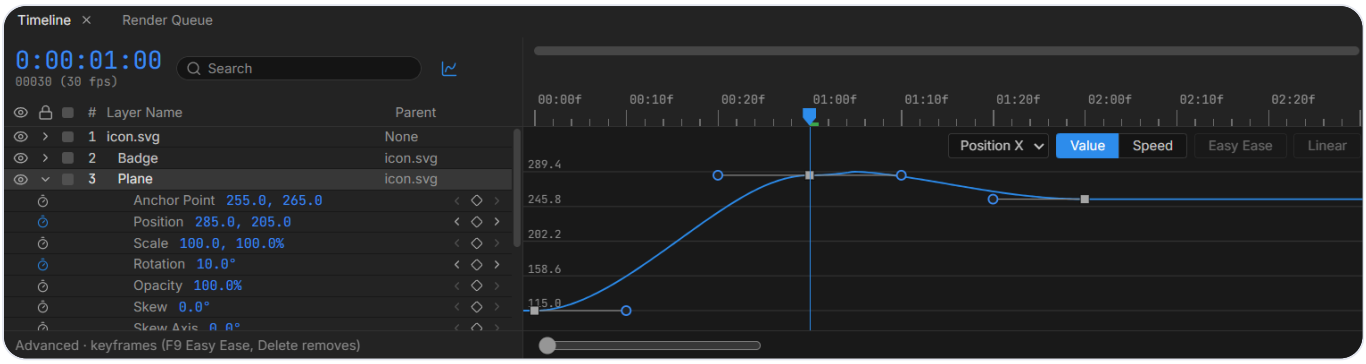
EASING	KEY	FEELS LIKE
Linear		Constant speed, mechanical.
Easy Ease	`F9`	Slows into and out of the keyframe: natural motion.
Easy Ease In	`Shift` + `F9`	Arrives gently (slows down into the keyframe).
Easy Ease Out	`Ctrl` + `Shift` + `F9`	Leaves gently (speeds up out of the keyframe).

For finer control, shape the curve in the Graph Editor.

11 Graph Editor

See and shape the timing of keyframes as curves.

Open it with `Shift` + `F3` or the graph button at the top of the timeline (Advanced workspace). Choose the property in the drop-down; only properties with keyframes are listed.

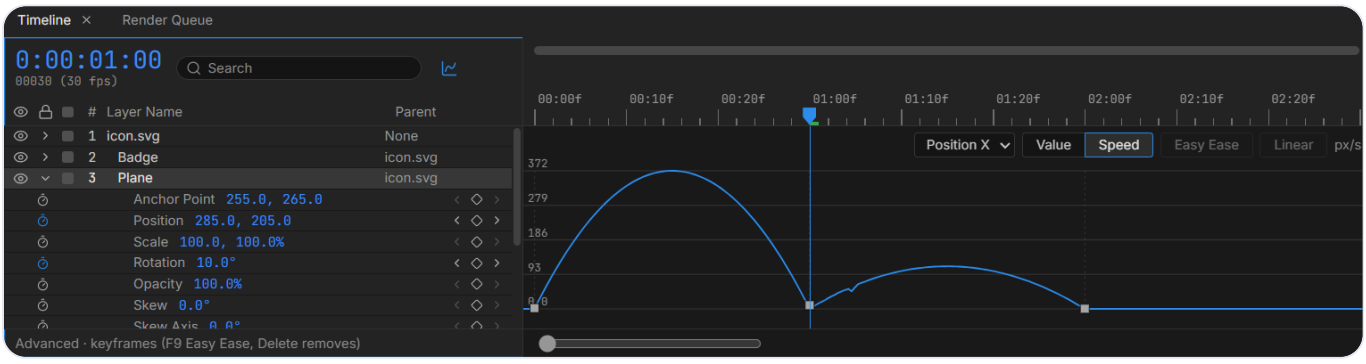


Value graph: the value of Position X over time, with keys and easing handles.

Value graph

- Drag a **key** (square) up or down to change its value, left or right to change its time.
- Drag the round **handles** to shape the easing: long, flat handles give a soft ease; short ones a sharp change.
- **Easy Ease** and **Linear** buttons set the selected segments.

Speed graph

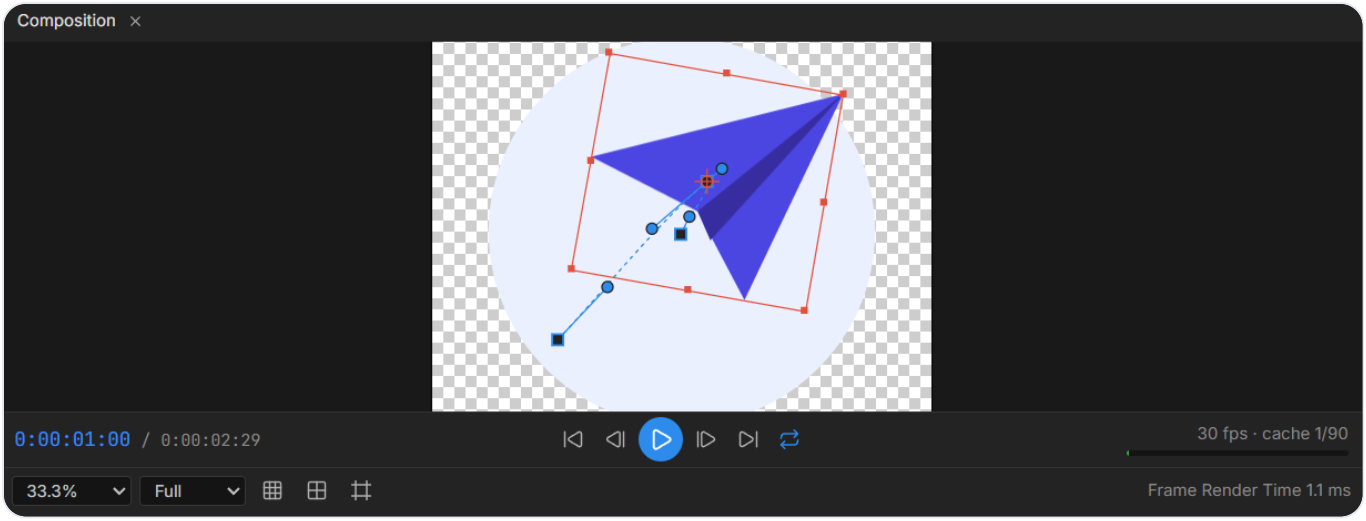


Speed graph of the plane: it speeds up and slows down between keys. For Position, it is the speed along the motion path.

The speed graph shows how fast the value changes (units per second). It is for reading: a smooth hill means a natural ease, a sharp corner means a jolt. Edit easing in the value graph and watch the result here.

12 Motion paths

Position keyframes form a path you can see and curve on the canvas.



The plane's motion path: squares are keyframes, circles are the curve handles.

- When Position has two or more keyframes, the selected layer shows its path on the canvas.
- Drag a **keyframe square** to move that position.
- Drag a **handle** to bend the path into a curve. Handles on both sides stay in line for a smooth curve.
- With the **Pen** tool `G`, click a keyframe to switch between a smooth curve and a sharp corner.
- Hold `Alt` while dragging a handle to break the pair and make a sharp turn.

> **TIP**
> Motion paths are exported to Lottie as real curves, so the web version moves exactly like the video.

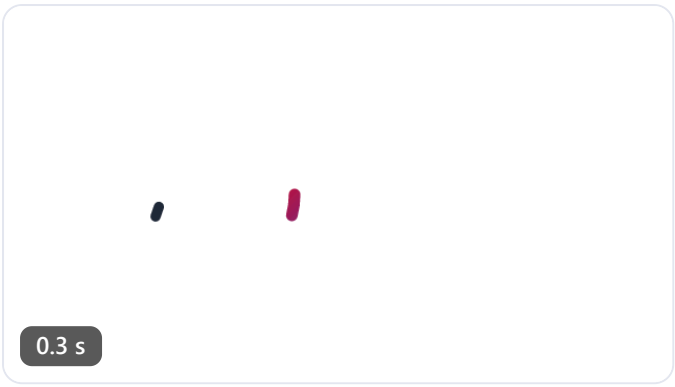
13 Trim Paths: drawing lines on

Show only part of each stroke to draw logos and line art on.



PROPERTY	MEANING
Start	Where the visible part begins, 0–100 % of the path length.
End	Where it ends. Animate End from 0 % to 100 % to draw the line on.
Offset	Shifts both along the path in degrees (360° = once around). Animate it to make a segment travel.

Trim Paths applies to every stroked path inside the layer, each path trimmed on its own. The easiest way is the *Draw On*, *Draw From Centre*, *Draw Off* and *Trace* presets; for full control, keyframe Start, End and Offset.



The ring and the wave of the logo sample are drawn on with Draw On.

> **IMPORTANT**
>
> Trim Paths only affects strokes. Filled shapes are not affected; reveal those with Wipe In or Iris In instead.

14 Effects and what each export supports

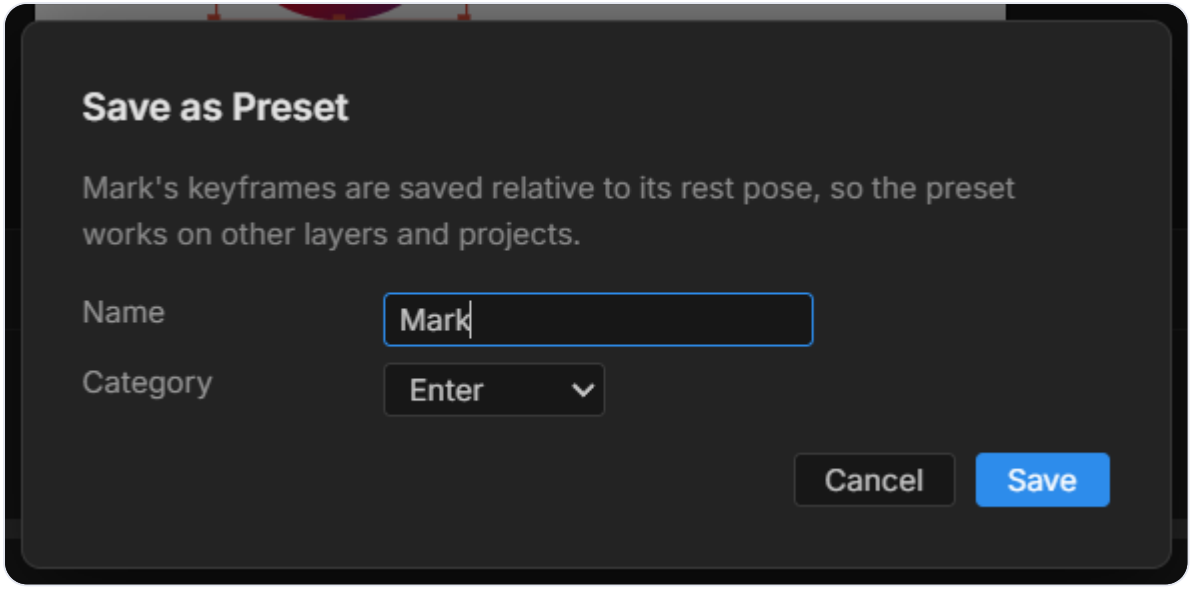
Every layer has Skew, Blur, Tint and Reveal in the Properties panel, each with its own stopwatch. Not every web format can show every effect, so plan with this table if you export to Lottie or SVG:

FEATURE	VIDEO	LOTTIE	ANIMATED SVG
Transform, opacity, skew	✓	✓	✓
Curved motion paths	✓	✓	✓
Trim Paths	✓	✓	✓
Blur	✓	✗	~ CSS blur (may differ slightly per browser)
Tint	✓	✓	✗
Reveal / wipe	✓	~ as a mask (not on the root SVG layer)	✗
Clip paths from the SVG	✓	~ outer clip only when nested	✓
SVG filters, patterns, blend modes	✓	✗	✓
Audio	✓	✗	✗

NOTE

Each export shows a compatibility report listing exactly what was simplified or left out, so nothing is lost silently.

15 Saving your own presets



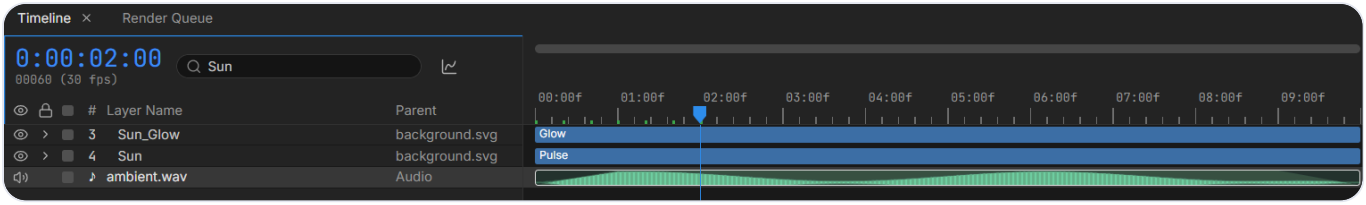
Save as Preset: keyframes are stored relative to the rest pose.

1. Animate a layer with keyframes until you like the motion.
2. Choose **Animation ▸ Save as Preset...**, give it a name and a category (Enter, Exit, Loop or Emphasis).
3. It appears under **My Presets** in the Presets panel, in every project. Apply it like any other preset.

Keyframes are saved as changes from the layer's rest pose (e.g. “move 200 px left and fade in”), so the preset works on other layers wherever they are. A project that uses your preset keeps its own copy, so it still opens on another computer. Delete a preset with the trash icon in My Presets.

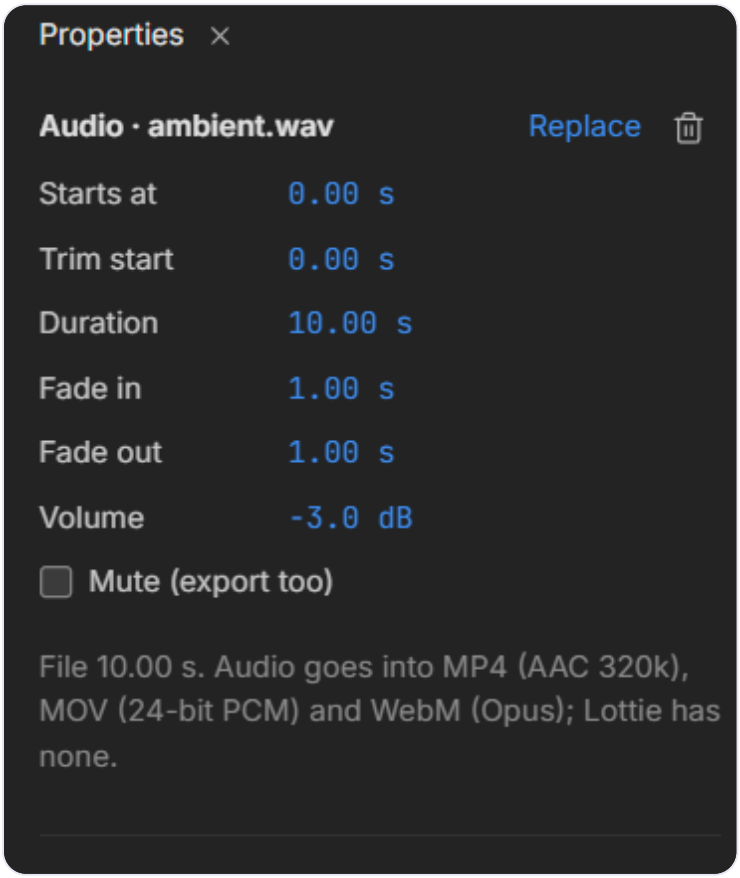
16 Audio

One soundtrack per project, with a waveform for timing.



The audio track at the bottom of the timeline, with its waveform and fades.

1. Choose **File** ▸ **Import Audio...** `Ctrl` + `Alt` + `I` or drag an audio file onto the window (WAV, MP3, AAC/M4A, FLAC, OGG).
2. The track appears at the bottom of the timeline. Drag it to move it in time; drag its edges to trim it.
3. Click the track to edit it in the Properties panel.



Audio properties.

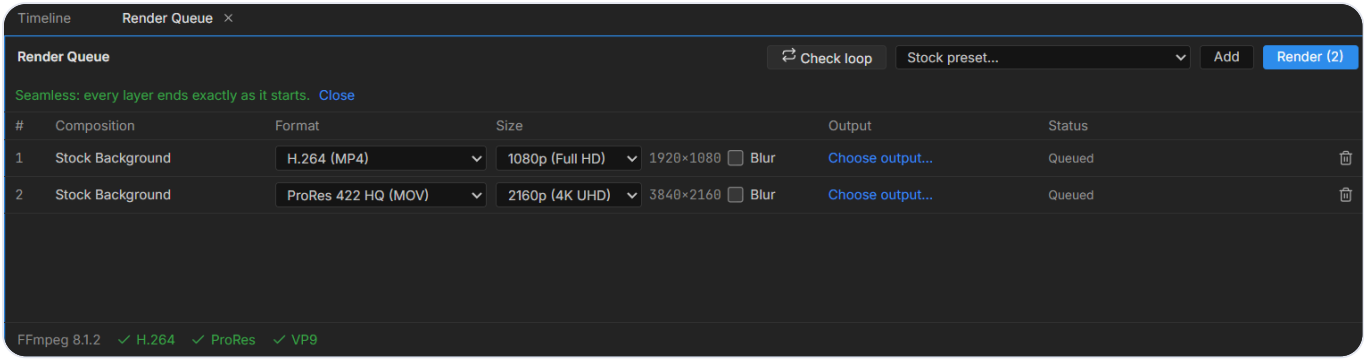
SETTING	MEANING
Starts at	Where the sound starts in the composition.
Trim start	Skips the beginning of the file.
Duration	How long it plays.
Fade in / Fade out	Smooth volume ramps at both ends (seconds).
Volume	Gain in dB (0 dB = original).
Mute	Silences the track, in preview and export.

Audio plays along with the preview and is mixed into MP4 (AAC 320 kbps), MOV (24-bit PCM) and WebM (Opus). Lottie and animated SVG have no sound. **Replace** swaps the file and keeps the timing; the trash icon removes the track.

> **TIP**
> The audio file is not stored inside the project. Keep it next to the `.motionator` file; if it is moved, the track shows *File missing* and **Replace** lets you find it again.

17 Video export: Render Queue

Render one composition to several formats and sizes in one go.



The Render Queue with two outputs of the background sample, after a loop check.

1. Click **Export** or press `Ctrl` + `M`: the composition is added to the queue. Add it again for more outputs.
2. Choose the **format** and **size** of each item, and tick **Blur** for motion blur.
3. Click **Choose output...** to set the file name.
4. Press **Render**. Items render one after another with progress and time left; each can be cancelled.

Formats

FORMAT	TRANSPARENCY	USE IT FOR
H.264 (MP4)	✗	Stock footage, social media, presentations: plays everywhere.
ProRes 422 HQ (MOV)	✗	Masters for editing and for stock sites that accept ProRes; large files.
ProRes 4444 (MOV)	✓	Overlays with transparency for video editors (Premiere, Resolve, Final Cut).
VP9 (WebM)	✓	Transparent video for websites.

Size scales the composition to 720p, 1080p, 1440p or 2160p (4K UHD) keeping its aspect ratio, or keeps the composition size. Because Motionator draws vectors, a 4K render is as sharp as the

composition at any size. H.264 and VP9 need even widths and heights; an odd size is shown in red.

Motion blur blends 8 sub-frames per frame (180° shutter) for natural blur on fast motion. It takes about 8× longer to render.

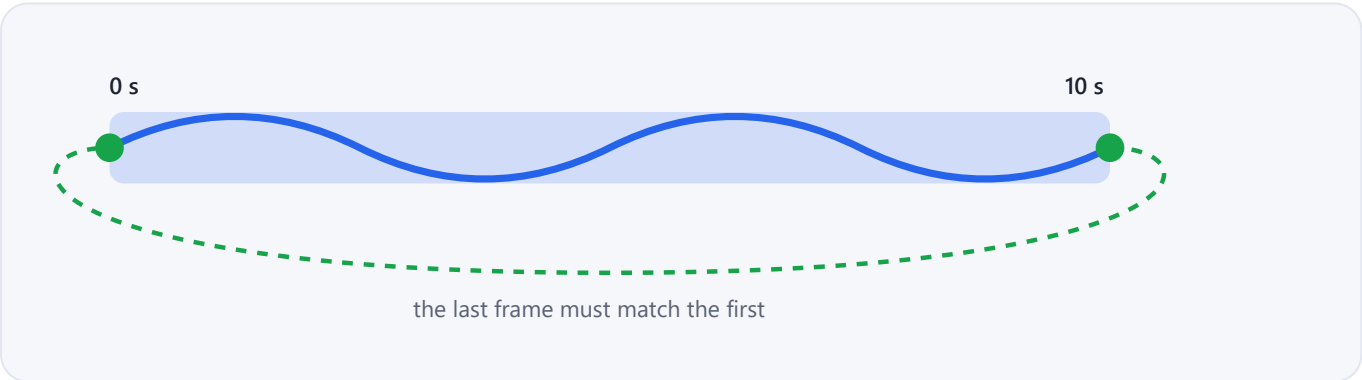
Stock presets and checks

Stock preset… adds an item already set up for microstock sites:

PRESET	OUTPUT
Microstock 4K · H.264	3840×2160, high-quality H.264 (CRF 12)
Microstock 4K · ProRes 422 HQ	3840×2160 master
Microstock HD · H.264	1920×1080, H.264 (CRF 14)

A yellow warning appears on an item when something is commonly rejected by stock sites: a clip shorter than 5 s or longer than 60 s, a height below 1080 px, a non-standard frame rate, or a format with alpha. Always check the current rules of the site you upload to.

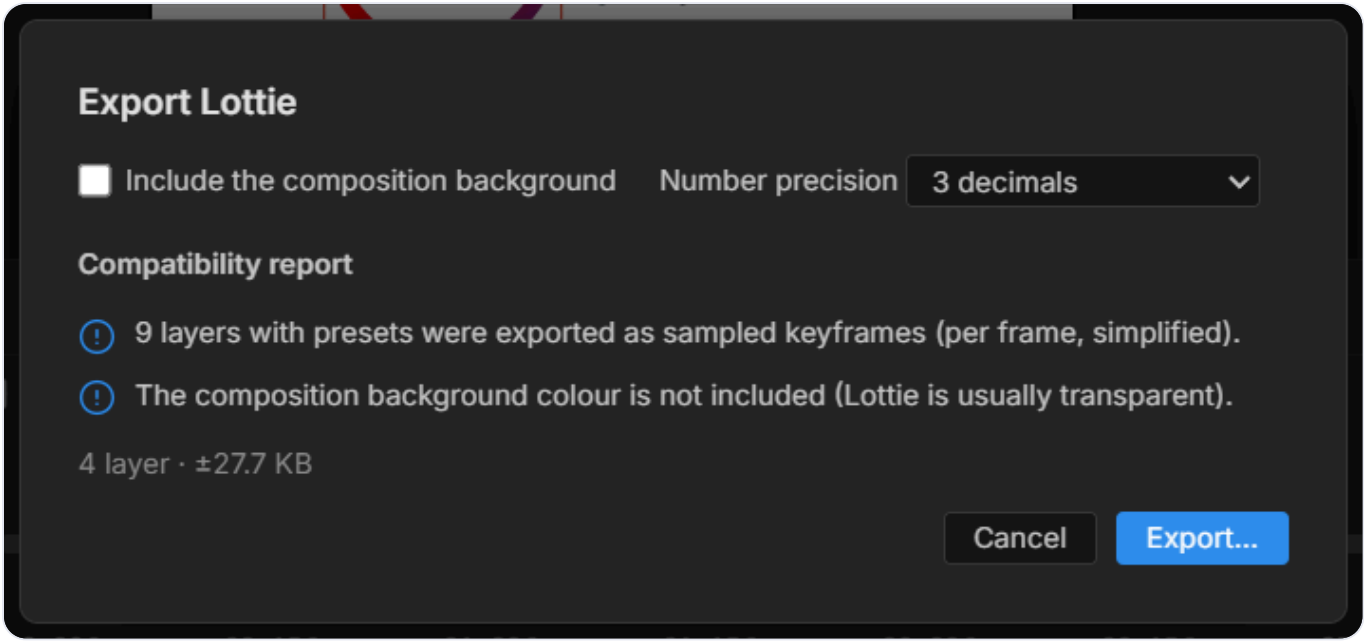
Checking a seamless loop



Looping backgrounds sell better when they repeat without a visible jump. **Check loop** compares every layer at the end of the composition with the start. If something differs, it lists the layers and properties. **Match the end to the start** adds keyframes at the end with the start values. Loop presets are always seamless; Enter, Exit and Emphasis presets cannot be fixed automatically, so use Loop presets or remove them.

18 Lottie export

Small JSON animations for websites and apps.



Export Lottie with its compatibility report and the estimated size.

1. Choose **File** ▸ **Export Lottie...** `Ctrl` + `Alt` + `L`.
2. Decide whether to include the composition background (Lottie is usually transparent) and the number precision (fewer decimals = smaller file).
3. Read the compatibility report, then click **Export...** and save the `.json` file.

Keyframes are exported as Lottie keyframes with their easing; presets are exported as sampled keyframes (simplified), which looks the same. Shapes, gradients, strokes, Trim Paths, clip paths and Tint are kept.

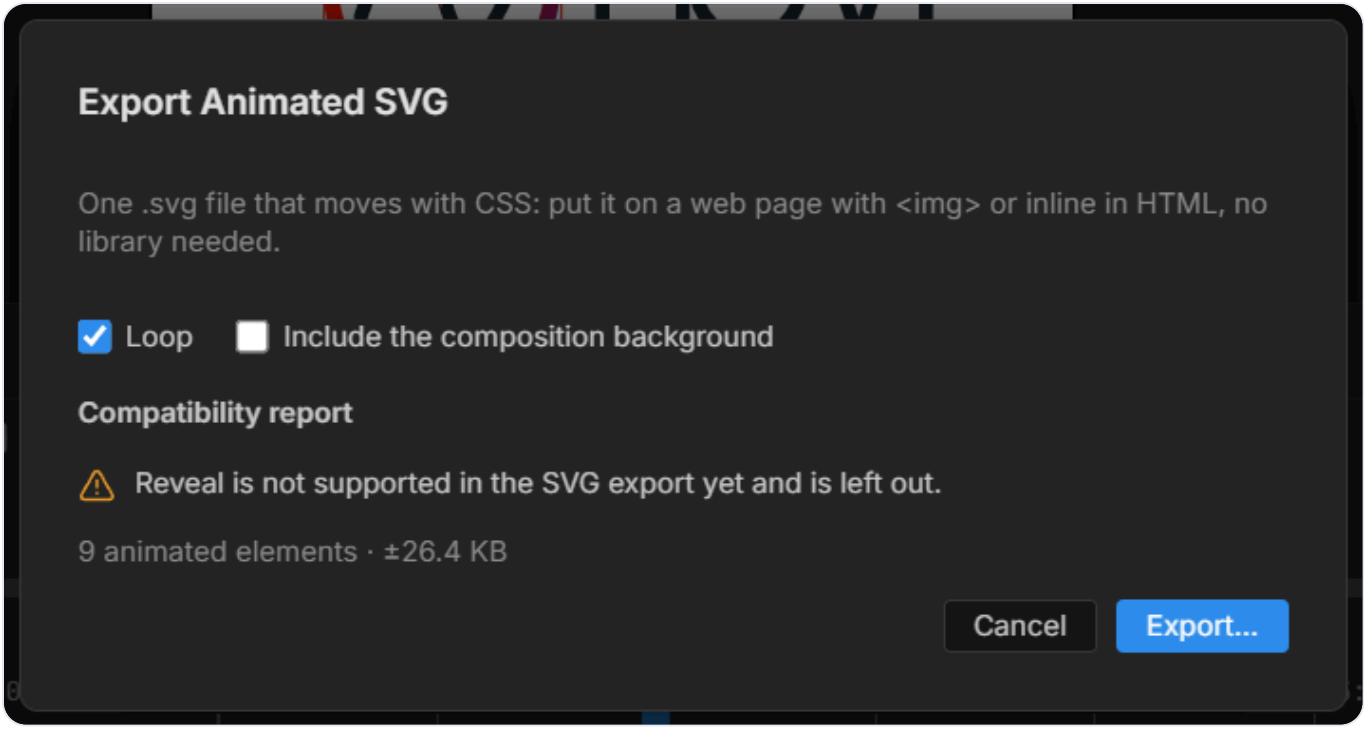
Using it on a web page

```
 <div id="logo" style="width:480px;height:270px"></div>
<script src="https://cdnjs.cloudflare.com/ajax/libs/lottie-web/5.12.2/lottie.min.js"></script>
<script>
  lottie.loadAnimation({
    container: document.getElementById("logo"),
    renderer: "svg",
    loop: true,
    autoplay: true,
    path: "logo.json"
  });
</script>
```

Or use the `<lottie-player>` / `<dotlottie-player>` web components, Lottie for Android/iOS, or upload the file to LottieFiles. Test the JSON in the player you will use: support for some features differs between players.

19 Animated SVG export

One self-contained .svg that moves with CSS, with no library needed.



Export Animated SVG: loop, background and the compatibility report.

Choose **File ▸ Export Animated SVG...** `Ctrl` + `Alt` + `G`. Tick **Loop** to repeat forever or leave it off to stop on the last frame. The file contains your artwork and CSS keyframe animations; it plays in every modern browser.

Putting it on a page

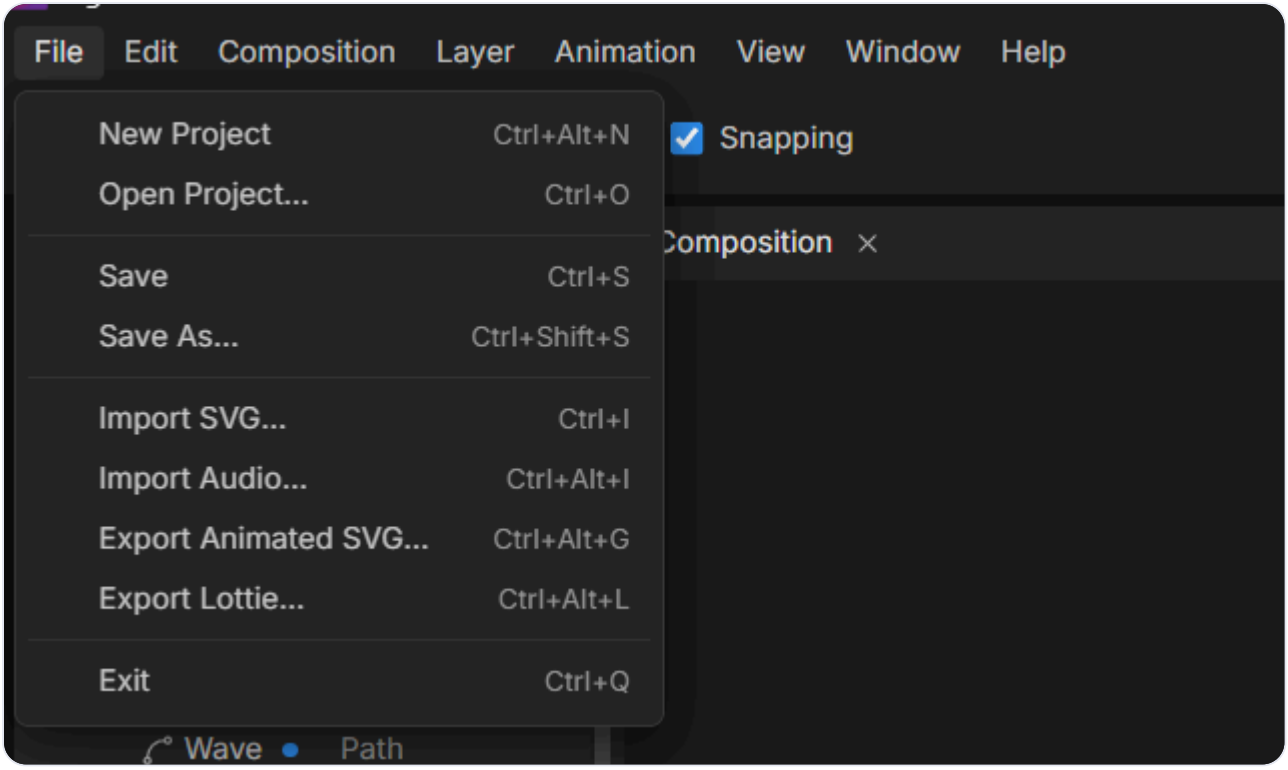
```
<!— as an image: simplest, the animation plays on its own —>
<img src="icon.svg" width="256" height="256" alt="Paper plane">

<!— inline in the HTML: lets your page's CSS reach inside —>
<div class="hero-icon">
  <!— paste the contents of icon.svg here —>
</div>
```

- **<img>** and CSS `background-image` work and keep the animation; scripts on the page cannot control it.
- Inline SVG can be paused with `animation-play-state: paused` on its elements.

CHOOSE	WHEN
Animated SVG	Icons, logos and illustrations on websites, with no JavaScript. Usually the smallest file.
Lottie	Apps (Android, iOS, Flutter, React Native), or when you need to control playback from code.
Video (VP9/MP4)	Complex scenes with heavy blur, many layers or filters.

20 Projects, auto-save and recovery



The File menu.

- **Save** `Ctrl` + `S` / **Save As...** `Ctrl` + `Shift` + `S` write a `.motionator` file. It contains the SVGs, keyframes, presets and your own presets used, so it is complete on its own (only audio stays a separate file).
- **New Project** `Ctrl` + `Alt` + `N` and **Open Project...** `Ctrl` + `O` ask first if there are unsaved changes. The title bar shows * when there are.
- **Auto-save** keeps a recovery copy every minute (change it in Preferences). It never overwrites your file.
- After a crash or power cut, Motionator offers **Recover project** on the next start. Choose Recover to continue, or Discard.

21 Tutorial: logo reveal with sound

Uses samples/logo.svg. Result: samples/logo.motionator.









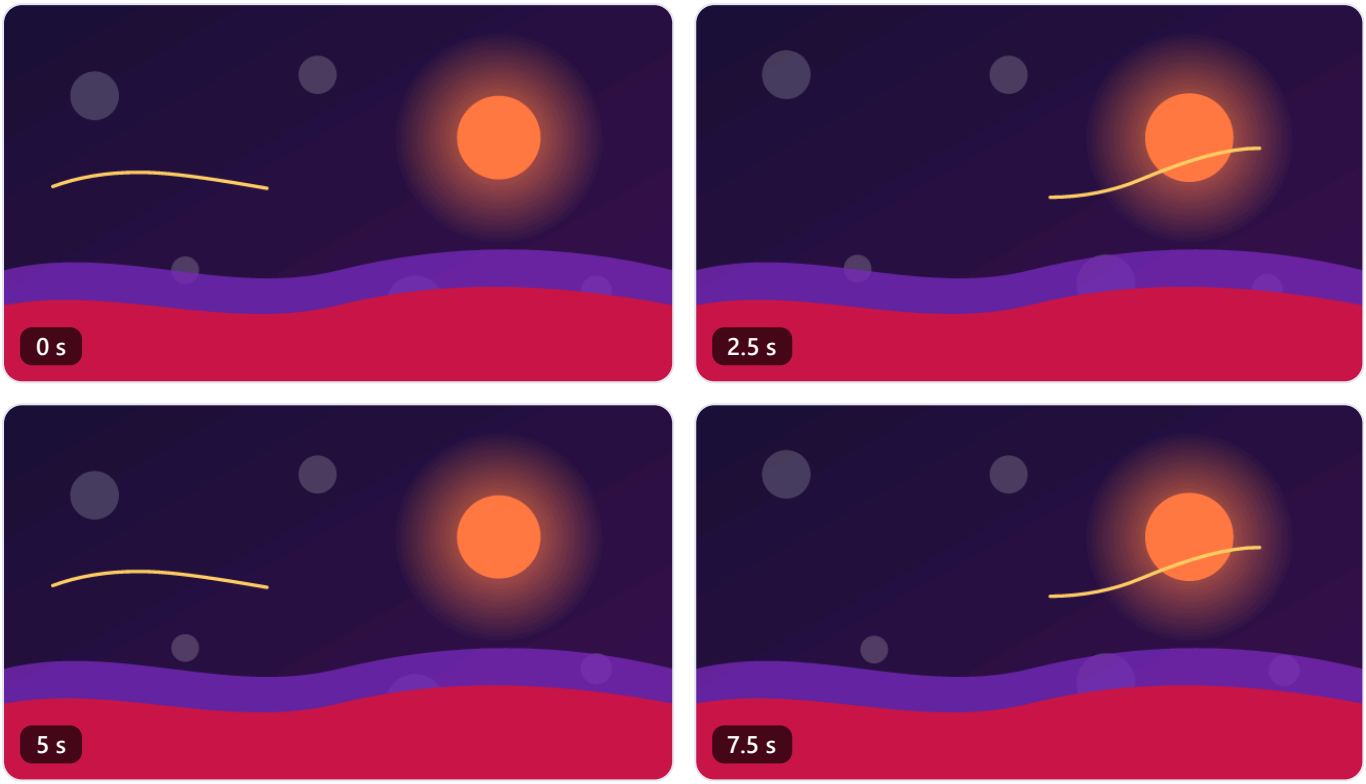
1. **New project.** Press `Ctrl` + `K`: 1920×1080, 30 fps, 4 s, white background.
2. **Import** `logo.svg` `Ctrl` + `I`. The layers are `Mark` (Ring, Wave, Dot), `Wordmark` (the letters) and `Underline`.
3. **Draw the mark.** At 0 s select `Ring` and apply *Draw On*. Select `Wave`, move the playhead to 0.3 s and apply *Draw On* again.
4. **Pop the dot.** At 0.9 s select `Dot` and apply *Elastic Pop*.
5. **Stagger the letters.** Tick *Stagger to child layers* (0.08 s, In order). At 1.0 s select `Wordmark` and apply *Pop Up*.
6. **Wipe the underline.** At 1.6 s select `Underline` and apply *Wipe In*; set *From* to the left.
7. **Accent.** At 3.0 s select `Mark` and apply *Pop*.
8. **Sound.** Import a short whoosh or chime with `Ctrl` + `Alt` + `I` and line it up with the Pop using the waveform.
9. **Export.** MP4 for social media; for the website, **Export Animated SVG** without Loop so it stops on the finished logo.

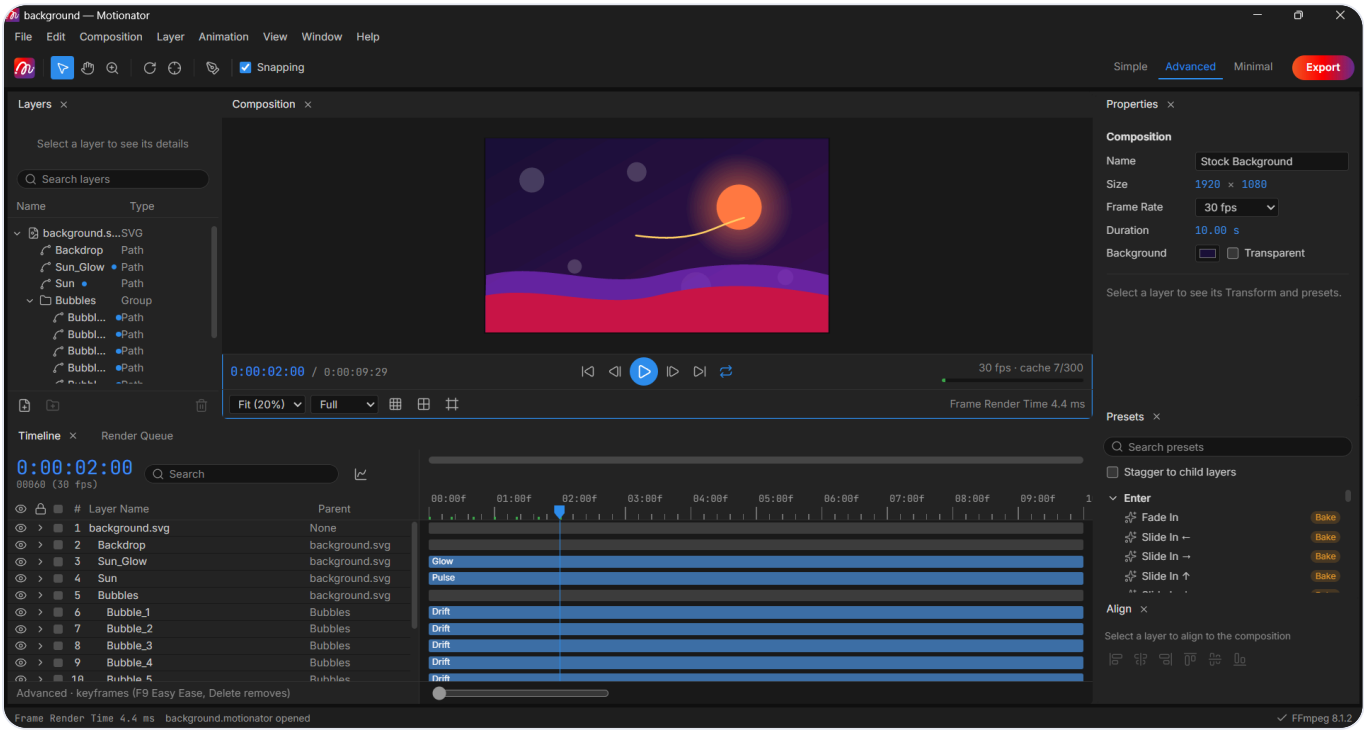
TIP

Rotate or scale a logo part around its centre? Check where its anchor point is first: press `Y` and drag it, or use Layer ▸ Center Anchor Point.

22 Tutorial: seamless 4K stock background

Uses samples/background.svg and ambient.wav. Result: samples/background.motionator.





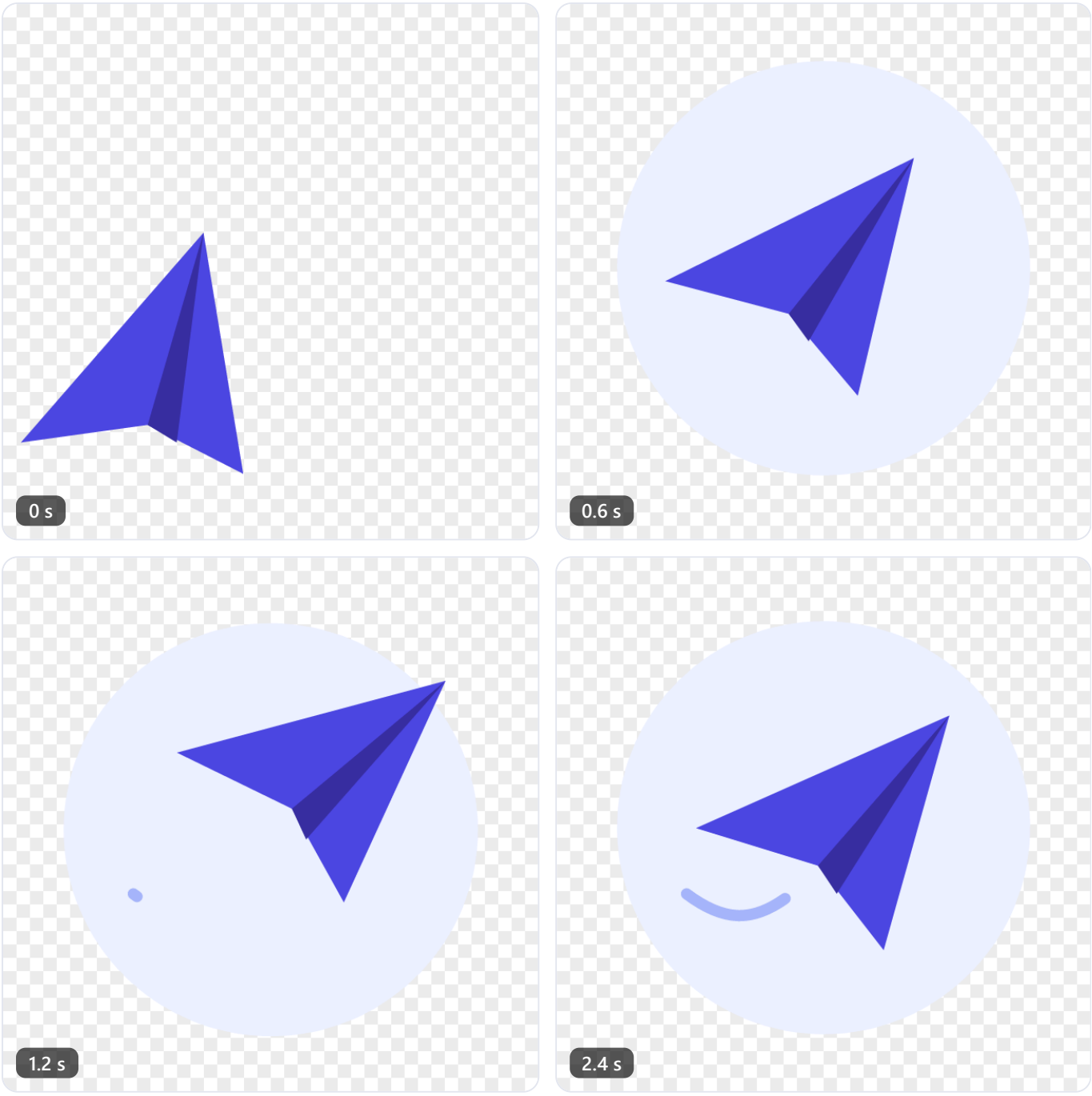
The background project: every layer uses Loop presets.

1. **Composition.** `Ctrl` + `K` : 3840×2160 (or 1920×1080, Motionator scales up sharply), 30 fps, **10 s**.
2. **Import** `background.svg` : Backdrop, Sun_Glow, Sun, Bubbles (Bubble_1–5), Wave_Back, Wave_Front, Ribbon.
3. **Use only Loop presets** so the end matches the start: *Glow* on Sun_Glow, *Pulse* on Sun, *Sway* on the waves, *Drift* with stagger on Bubbles, *Trace* on Ribbon.
4. **Make periods divide the duration.** In a 10 s clip, use periods of 10, 5, 2.5 or 2 s so every cycle completes exactly at the end.
5. **Audio (optional).** Import `ambient.wav` ; set Fade in and Fade out to 1 s. Many stock sites prefer clips without sound, so you can also mute it for the upload.
6. **Check the loop.** `Ctrl` + `M` , then **Check loop** in the Render Queue. It should say *Seamless*.
7. **Render.** Choose *Microstock 4K · H.264* and, if the site accepts it, *Microstock 4K · ProRes 422 HQ*. Fix any yellow warning, then Render.

> **NOTE**
> Upload checklist: 5–60 s long, at least 1920×1080, standard frame rate, no logos or trademarks you do not own, a descriptive title and keywords. Mention “seamless loop” in the description.

23 Tutorial: animated icon for a website

Uses samples/icon.svg. Result: samples/icon.motionator.



1. **Composition** 512×512, 30 fps, 3 s, **Transparent** ticked.
2. **Import** `icon.svg` : Badge, Plane (Plane_Body, Plane_Fold) and Trail.
3. **Fly the plane with keyframes.** Select `Plane` . At 0 s, move it to the lower left and turn on the Position and Rotation stopwatches. At 1 s, move it to the centre; at 2 s, slightly up and right.

4. **Curve the path.** Drag the handles of the motion path on the canvas so the plane swoops.
5. **Ease it.** Select all keyframes and press `F9`. Check the speed graph `Shift` + `F3`: smooth hills, no sharp corners.
6. **Trail and badge.** *Draw On* on `Trail` at 0.2 s, *Scale Pop* on `Badge` at 0 s, and *Hover* (Loop) on Plane for life after it lands.
7. **Export** both **Lottie** `Ctrl` + `Alt` + `L` for the app and **Animated SVG** `Ctrl` + `Alt` + `G` for the website, and compare the sizes in the dialogs.

24 Keyboard shortcuts

Shortcuts follow After Effects where possible.

File

COMMAND	SHORTCUT
New Project	`Ctrl` + `Alt` + `N`
Open Project...	`Ctrl` + `O`
Save	`Ctrl` + `S`
Save As...	`Ctrl` + `Shift` + `S`
Import SVG...	`Ctrl` + `I`
Import Audio...	`Ctrl` + `Alt` + `I`
Export Animated SVG...	`Ctrl` + `Alt` + `G`
Export Lottie...	`Ctrl` + `Alt` + `L`
Exit	`Ctrl` + `Q`

Edit

COMMAND	SHORTCUT
Undo	`Ctrl` + `Z`
Redo	`Ctrl` + `Shift` + `Z` `Ctrl` + `Y`
Delete	`Delete`
Deselect All	`Ctrl` + `Shift` + `A`
Preferences...	`Ctrl` + `Alt` + `;`

Composition and layer

COMMAND	SHORTCUT
Composition Settings...	`Ctrl` + `K`
Add to Render Queue	`Ctrl` + `M`
Center Anchor Point	`Ctrl` + `Alt` + `Home`

Animation

COMMAND	SHORTCUT
Easy Ease	`F9`
Easy Ease In	`Shift` + `F9`
Easy Ease Out	`Ctrl` + `Shift` + `F9`
Previous Keyframe	`J`
Next Keyframe	`K`
Reveal Animated Properties	`U`
Graph Editor	`Shift` + `F3`

Time and playback

COMMAND	SHORTCUT
Play / Pause	`Space`
Previous Frame	`Page Up` `←`
Next Frame	`Page Down` `→`
Back 10 Frames	`Shift` + `Page Up` `Shift` + `←`

COMMAND	SHORTCUT
Forward 10 Frames	`Shift` + `Page Down` `Shift` + `→`
Go to Start	`Home`
Go to End	`End`

View

COMMAND	SHORTCUT
Grid	`Ctrl` + `'`
Snapping	`Ctrl` + `Shift` + `;`
Zoom In	`.`
Zoom Out	`,`
Actual Size (100%)	`/`
Fit to Panel	`Shift` + `/`

Workspaces

COMMAND	SHORTCUT
Workspace: Simple	`Shift` + `F10`
Workspace: Advanced	`Shift` + `F11`
Workspace: Minimal	`Shift` + `F12`

Tools

COMMAND	SHORTCUT
Selection Tool	`V`
Hand Tool	`H`
Zoom Tool	`Z`
Rotation Tool	`W`
Anchor Point Tool	`Y`
Pen Tool (motion path)	`G`

MOUSE	ACTION
Space + drag	Pan the canvas
Ctrl/Alt + wheel	Zoom the canvas around the pointer
Shift + drag	Move only horizontally/vertically; rotate in 45° steps
Double-click a preset	Apply it to the selected layer
Esc while dragging a panel	Cancel the move

25 Troubleshooting and FAQ

My SVG imports as one layer.

The elements have no ids. Name the layers and groups in your design tool and export with ids (see Preparing SVG files).

Text is missing.

Convert text to outlines before exporting the SVG.

Draw On does nothing.

The paths have no stroke. Trim Paths only works on strokes; use Wipe In for filled shapes.

Rotation swings around the wrong point.

Move the anchor point with the Anchor Point tool `Y` or centre it with `Ctrl` + `Alt` + `Home`.

A preset seems to have no effect.

Check the playhead: Enter presets start there, so the motion may be later in the timeline. Another Enter/Exit preset on the same property may also override it.

Playback stutters.

Let the cache fill (green bar), lower the preview resolution to Half, or switch off blur while working.

Export failed / FFmpeg not found.

Reinstall Motionator. Also check that the output folder is writable and the disk has space (ProRes 4K needs several GB per minute).

“Width and height must be even”.

Pick a size preset or set an even composition size; H.264 and VP9 need even sizes.

The Lottie looks different in my app.

Read the compatibility report; blur, filters and blend modes are not part of Lottie. Players also differ: test in the one you ship.

A panel disappeared.

Open it from the Window menu, or use Window ▸ Reset Layout.

The audio track says File missing.

The audio was moved. Click the track and use Replace to locate it.

26 About and licences

Motionator is built with Tauri, React and Rust; vector rendering uses resvg/tiny-skia. **Help ▸ About** lists the version and the open-source components.

FFmpeg is included as a separate program under the GPL v3 and is used only to encode video and audio. Your exported videos, Lottie files and SVGs are yours to use commercially, including on stock sites.

The sample files in `samples/` were made for this guide and may be used freely for practice.
